Computer Vision with Deep Learning

Lecture 15: Light and shading- II

slides credit: Ioannis Gkioulekas, Lana Lazebnik, Raymond Yeh, David Lindell, (and whoever mentioned in specific slides)

Announcements

- HW2 grading in progress

- Project proposal due March 23rd
  - Online Q&A this Thursday from 17:00-17:45 on Zoom

Previously On

Computer Vision with Deep Learning

how do capture an image?

... and an object we like to photograph

real-world object

digital sensor (CCD or CMOS)

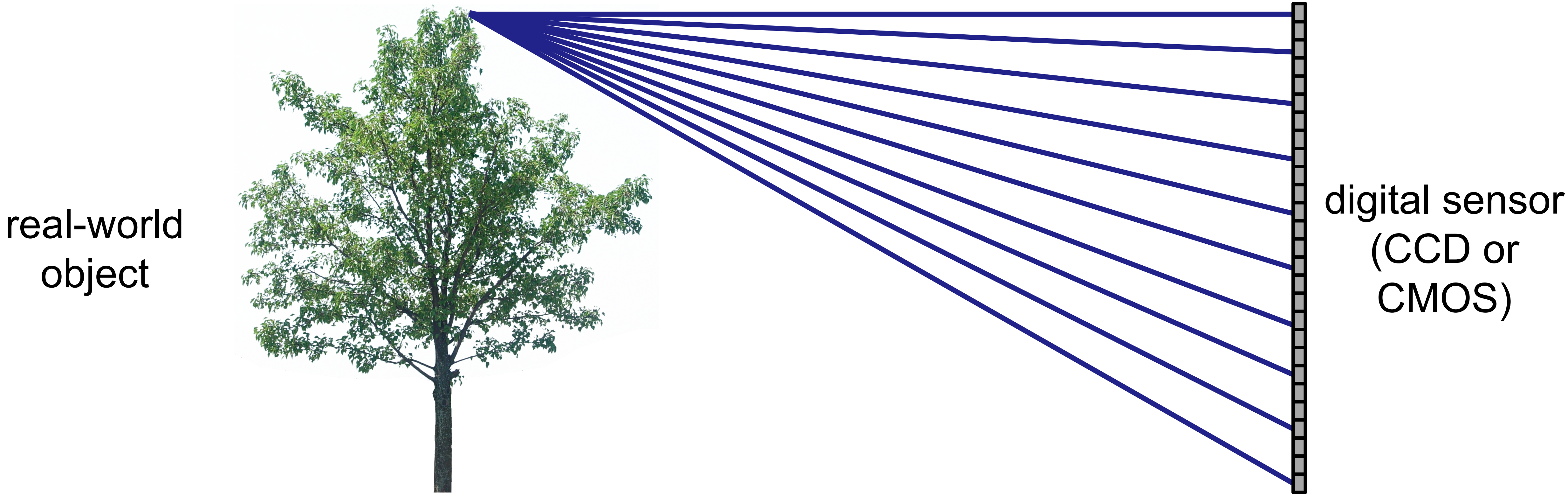
What would an image taken like this look like?

Bare-sensor imaging

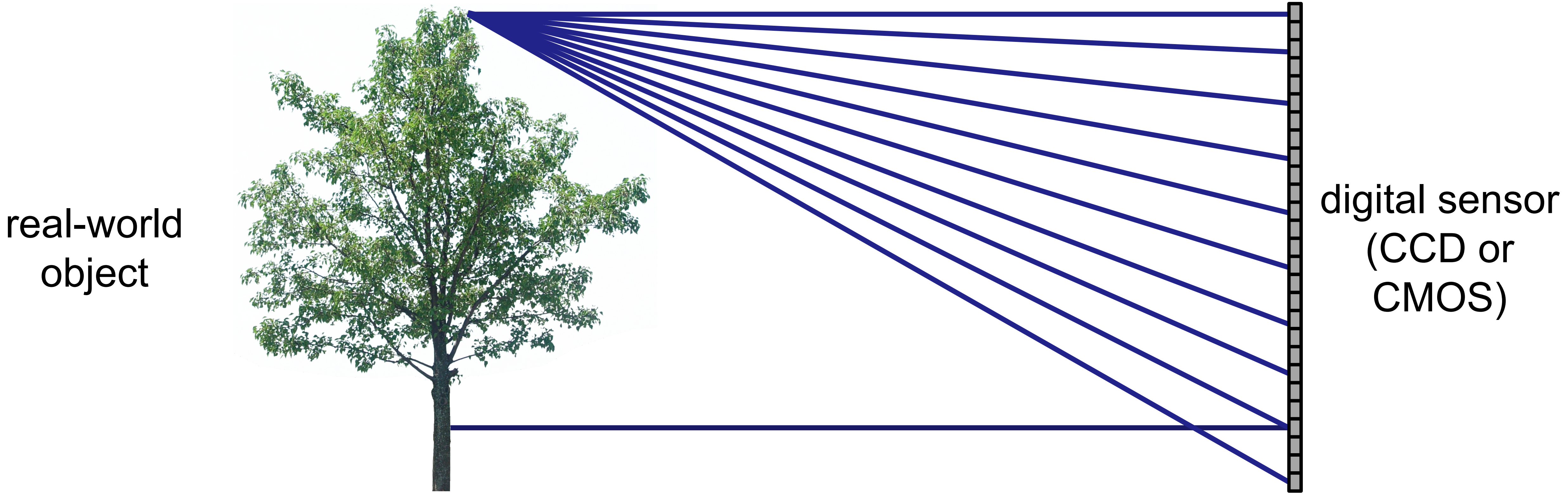
real-world object

digital sensor (CCD or CMOS)

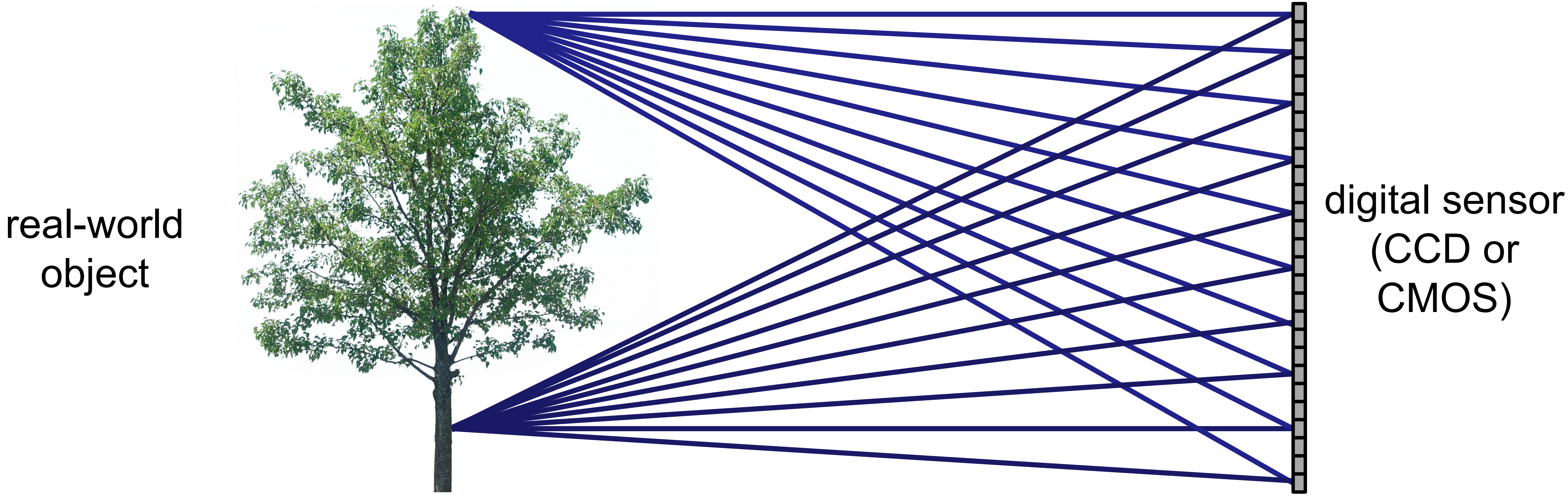
Bare-sensor imaging



Bare-sensor imaging



Bare-sensor imaging



All scene points contribute to all sensor pixels

What does the image on the sensor look like?

Bare-sensor imaging

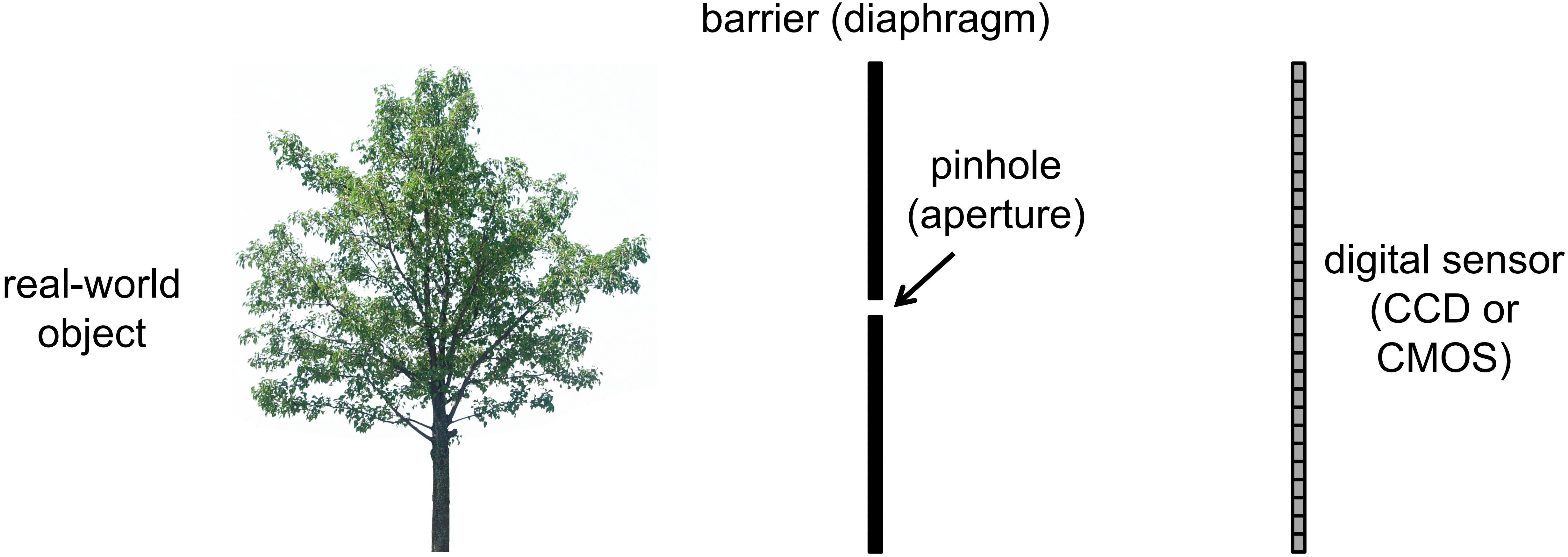
All scene points contribute to all sensor pixels

What can we do to make our image look better?

real-world
object

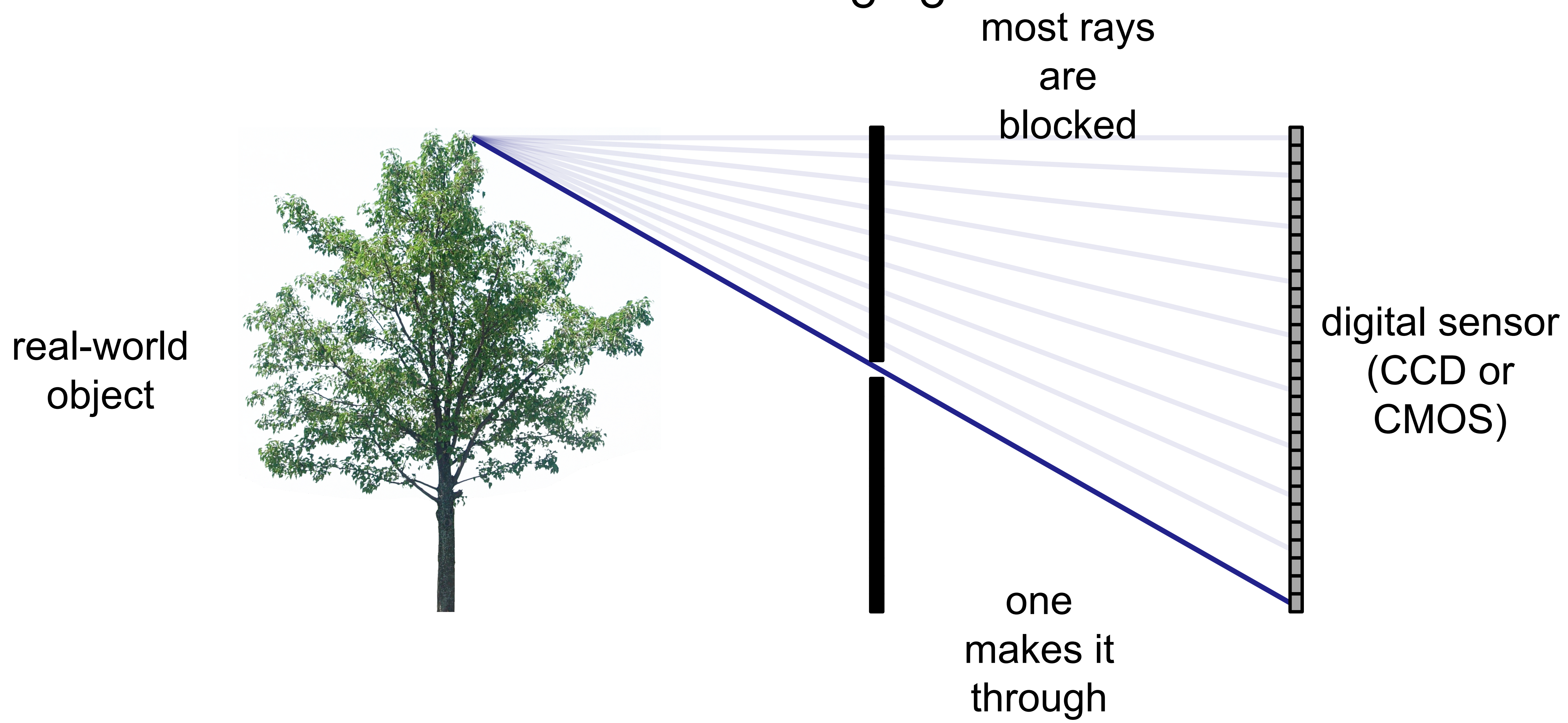
digital sensor
(CCD or
CMOS)

Let's add something to this scene

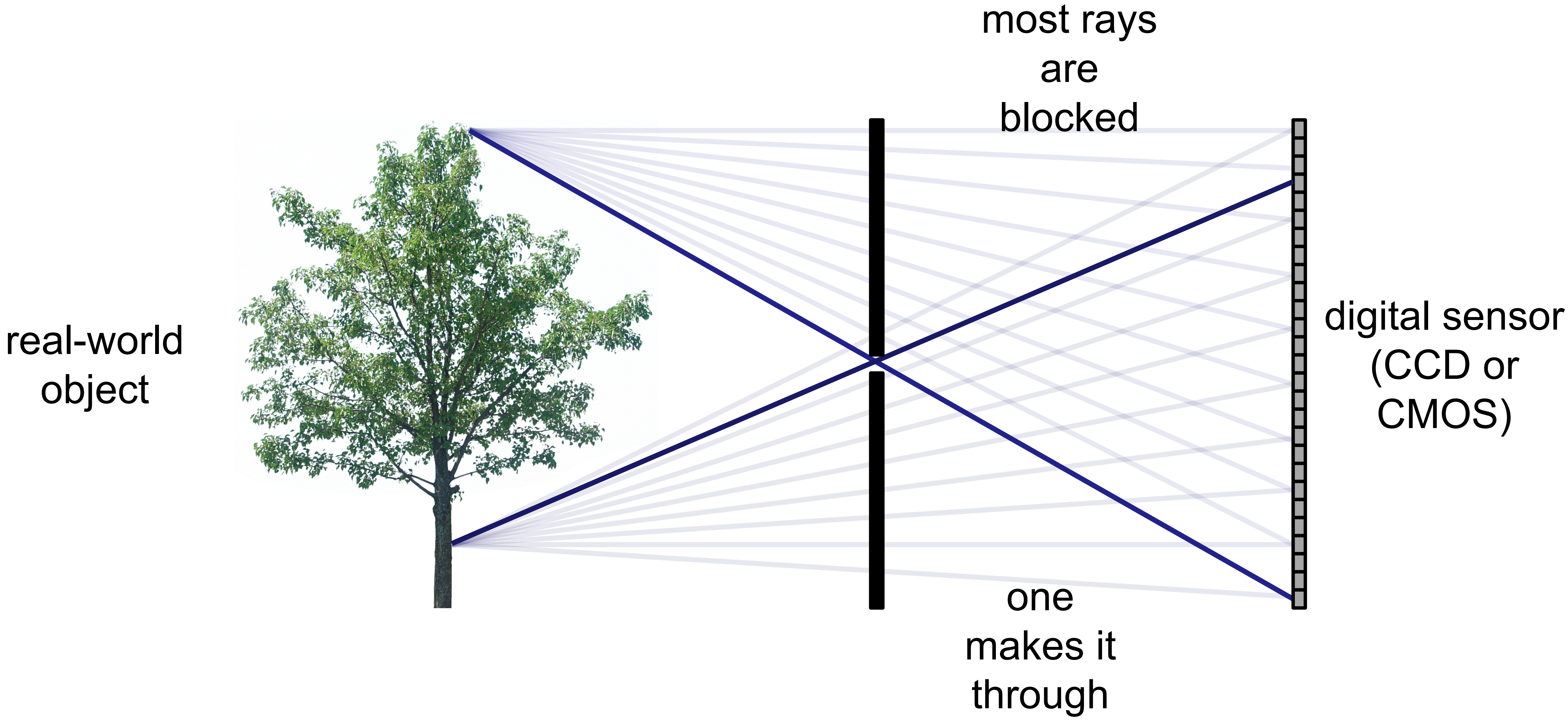


What would an image taken like this look like?

Pinhole imaging

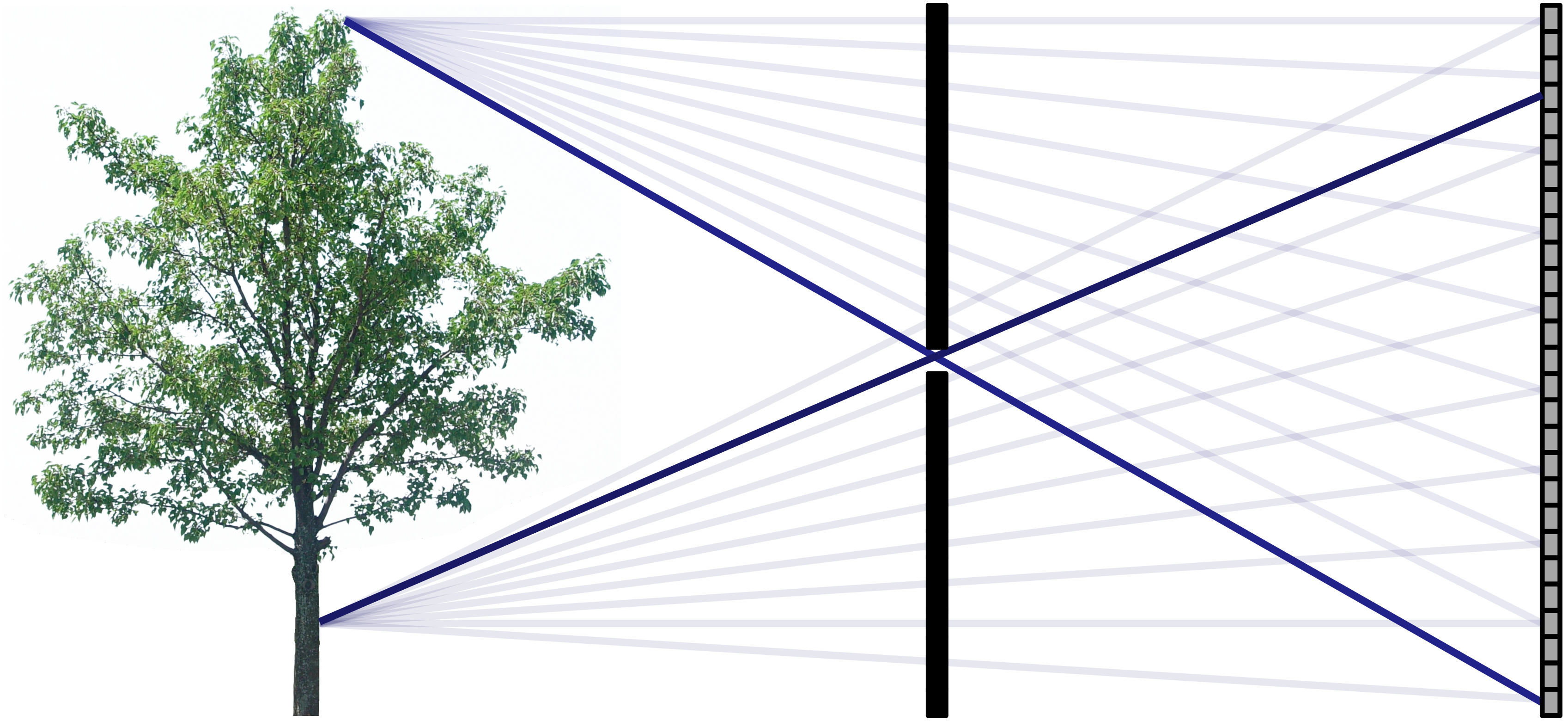


Pinhole imaging



Pinhole imaging

real-world object

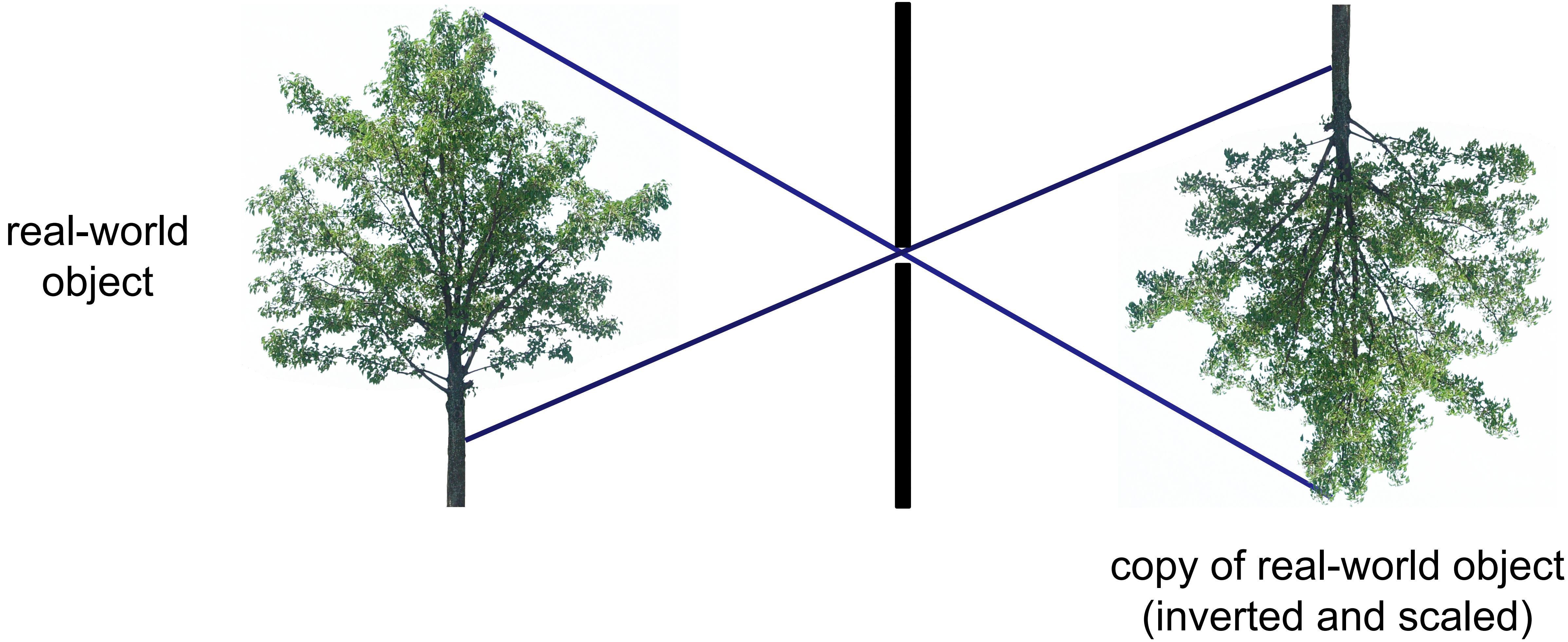


digital sensor (CCD or CMOS)

What does the image on the sensor look like?

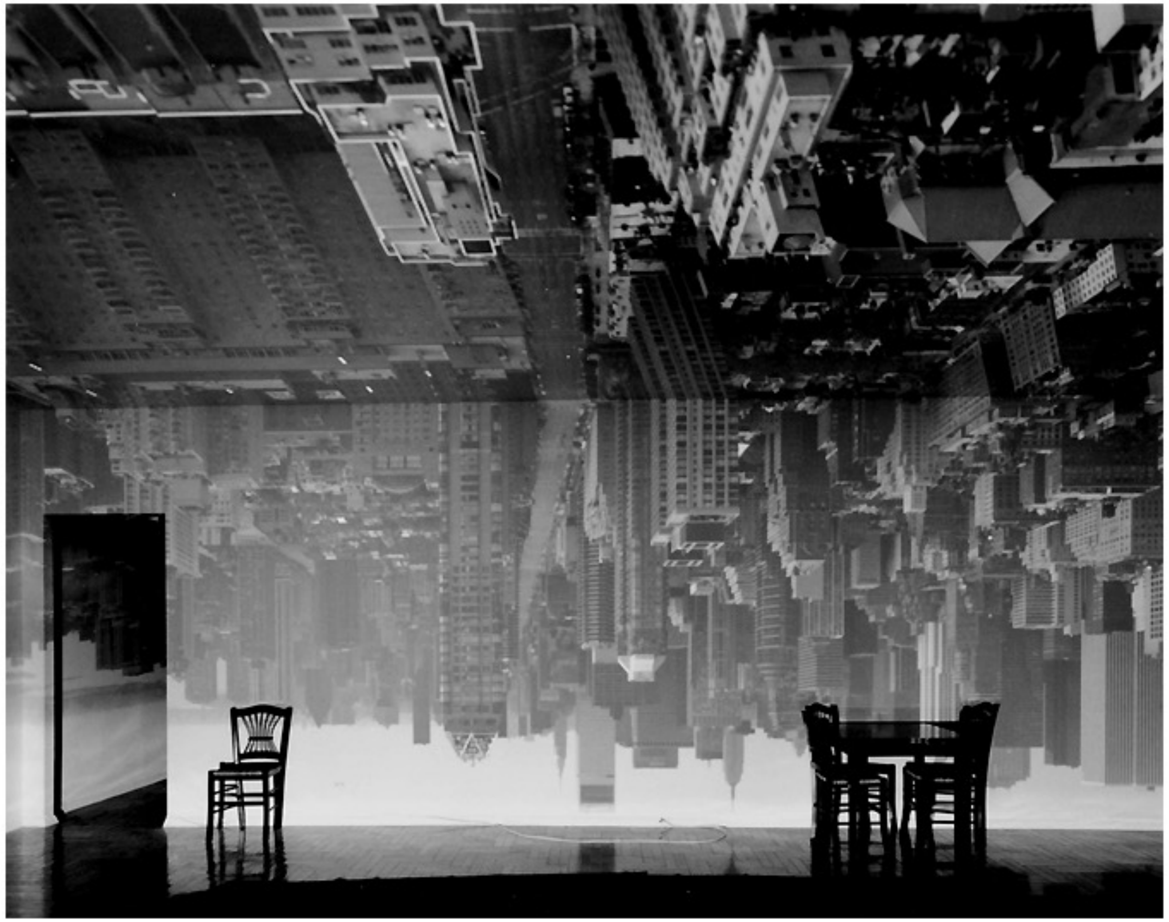
Each scene point contributes to only one sensor pixel

Pinhole imaging



Accidental pinholes

Turning a room into a camera obscura



Abelardo Morell, Camera Obscura Image of Manhattan View Looking South in Large Room, 1996

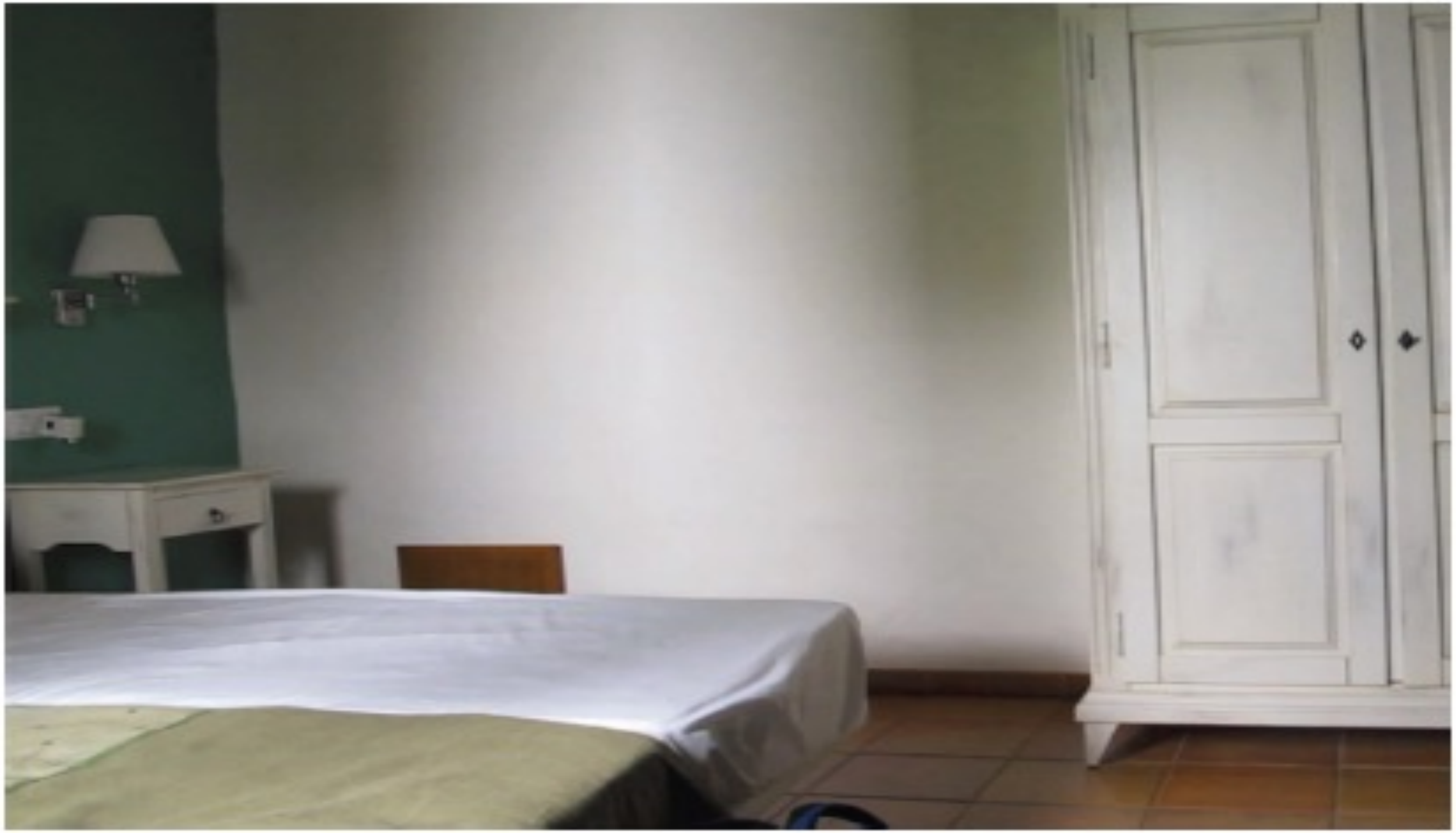
After scouting rooms and reserving one for at least a day, Morell masks the windows except for the aperture. He controls three elements: the size of the hole, with a smaller one yielding a sharper but dimmer image; the length of the exposure, usually eight hours; and the distance from the hole to the surface on which the outside image falls and which he will photograph. He used 4 x 5 and 8 x 10 view cameras and lenses ranging from 75 to 150 mm.

After he's done inside, it gets harder. "I leave the room and I am constantly checking the weather, I'm hoping the maid reads my note not to come in, I'm worrying that the sun will hit the plastic masking and it will fall down, or that I didn't trigger the lens."

From *Grand Images Through a Tiny Opening*, **Photo District News,** February 2005

https://www.abelardomorell.net/camera-obscura

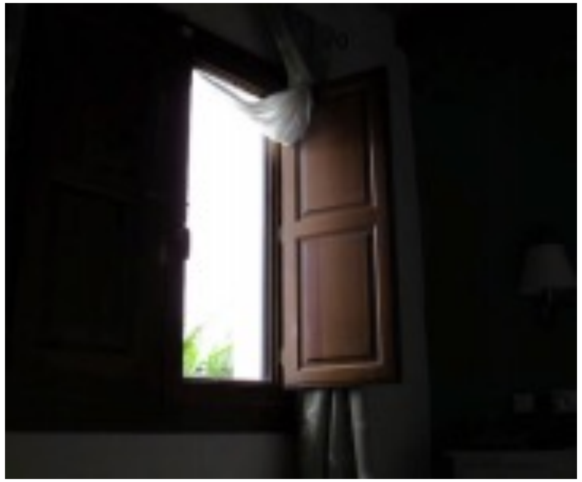
What does this image say about the world outside?



Accidental pinhole camera

Antonio Torralba, William T. Freeman
Computer Science and Artificial Intelligence Laboratory (CSAIL)
MIT
torralba@mit.edu, billf@mit.edu

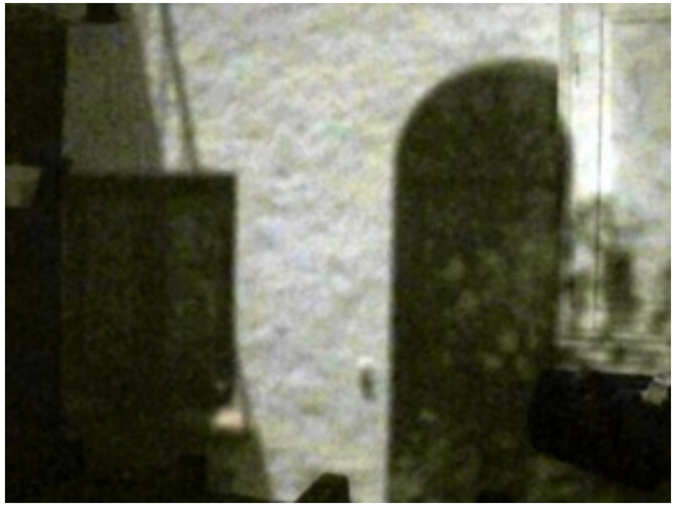
Accidental pinhole camera



window is an aperture

projected pattern on the wall

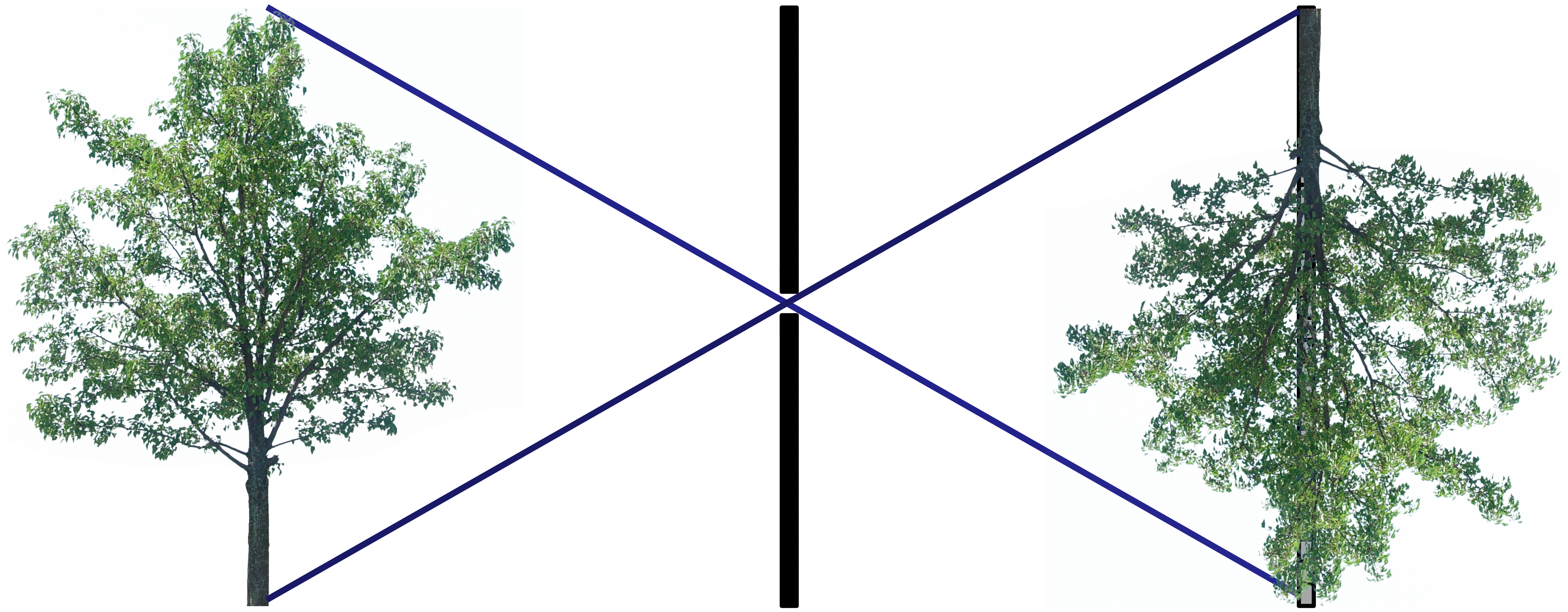
upside down



window with smaller gap

view outside window

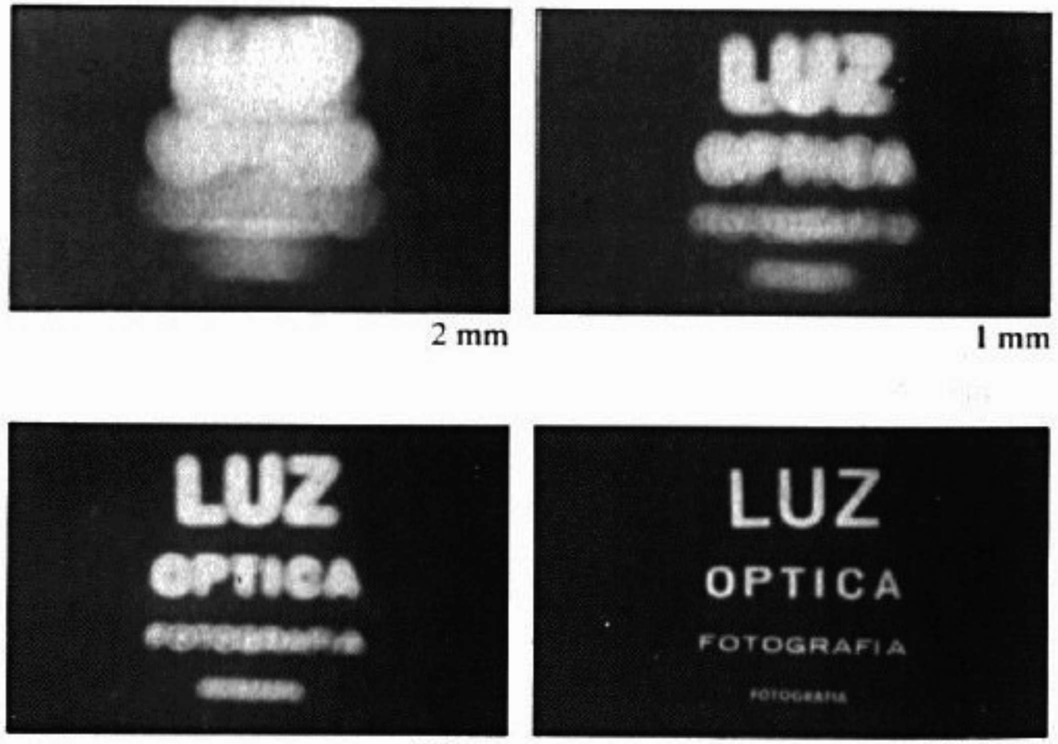
Pinhole camera trade-off



Small (ideal) pinhole:

1. Image is sharp.
2. Signal-to-noise ratio is low.

Shrinking the aperture



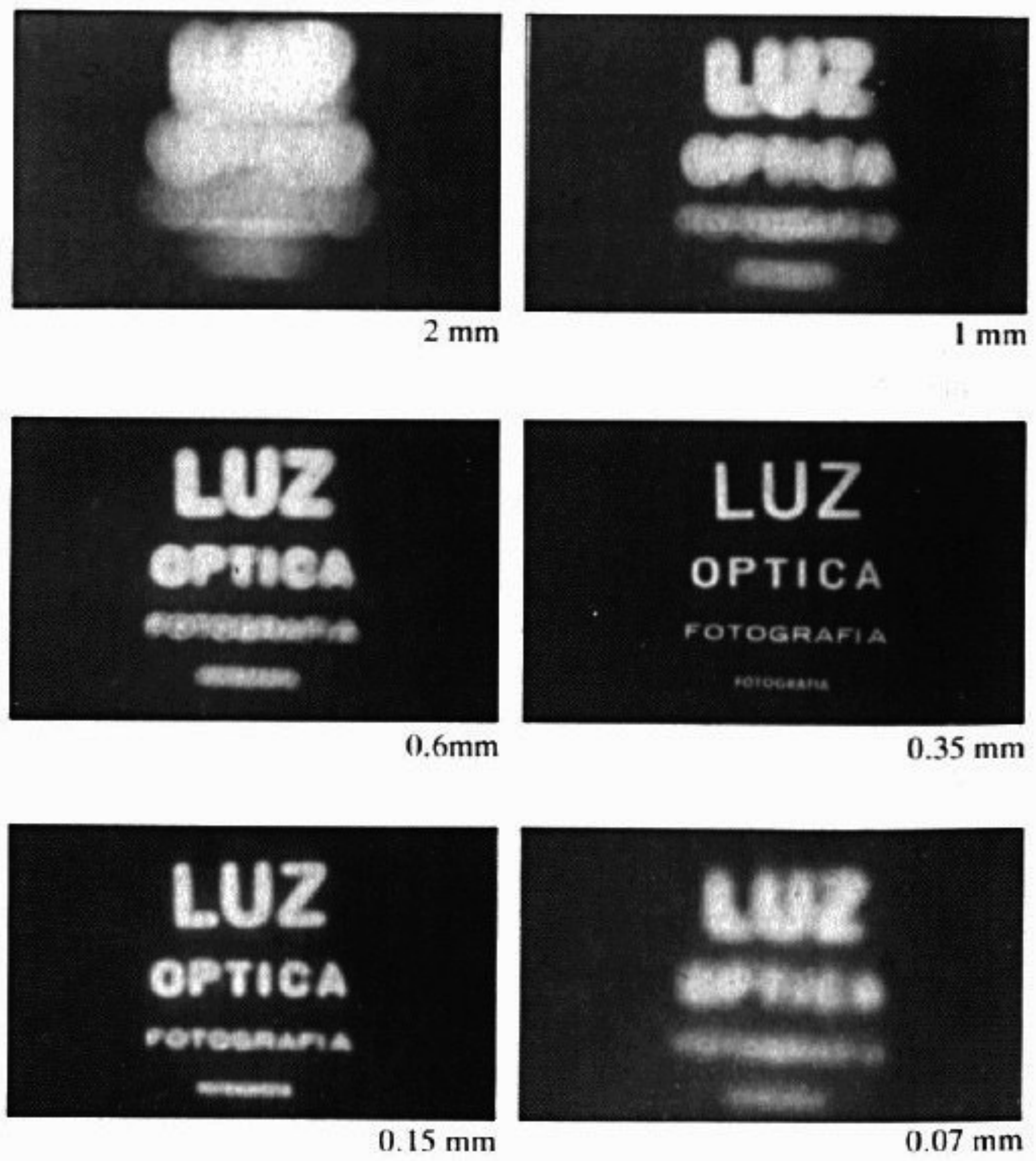
2 mm 1 mm 0.6mm 0.35 mm

Why not make the aperture as small as possible?

- Less light gets through
- Diffraction!

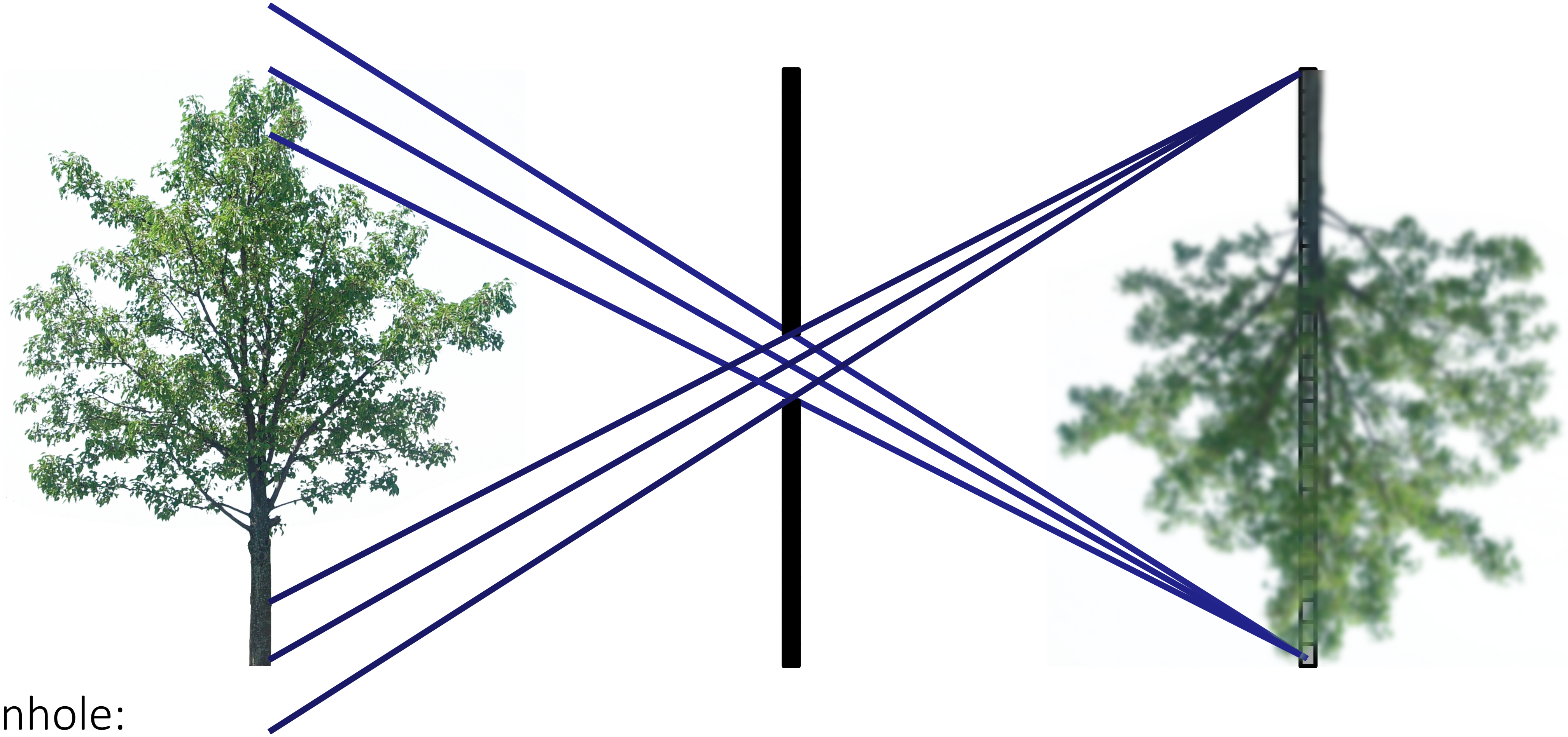
Slide by Steve Seitz

Shrinking the aperture



diffraction

Pinhole camera trade-off

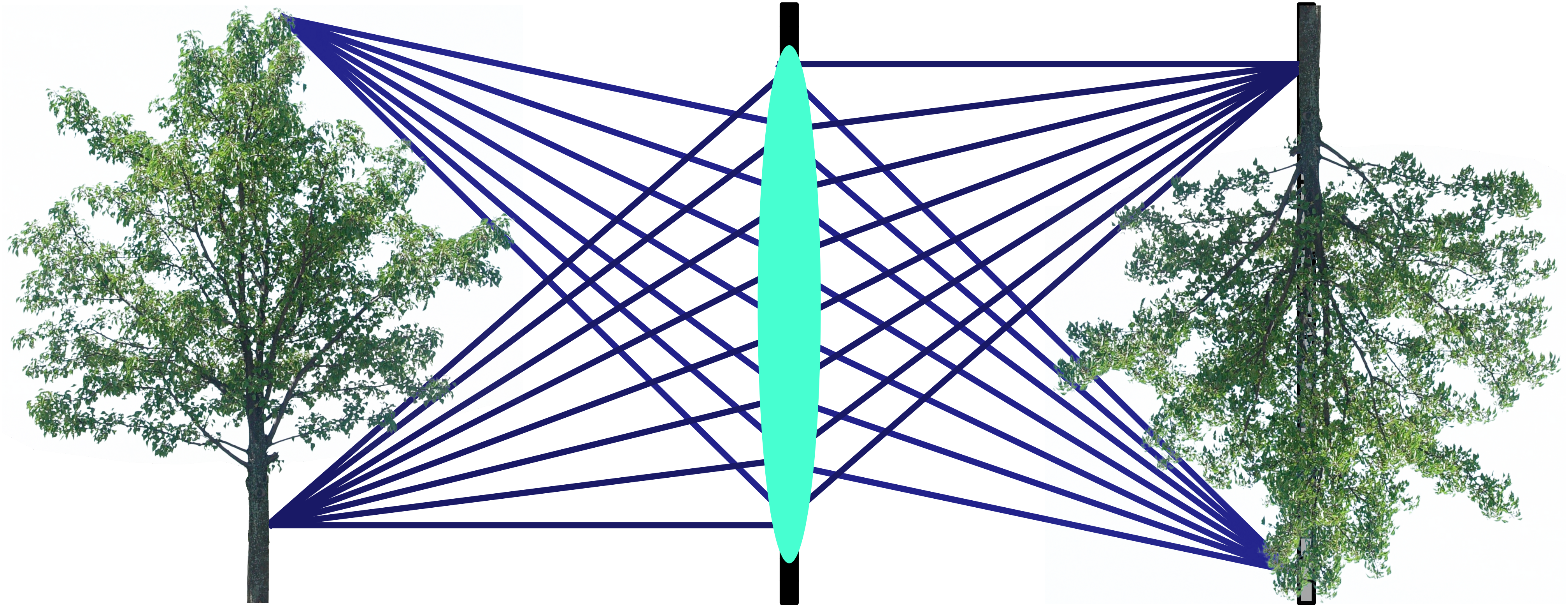


Large pinhole:

1. Image is blurry.
2. Signal-to-noise ratio is high.

Can we get best of both worlds?

Almost, by using lenses

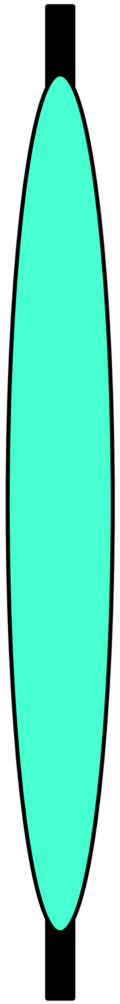


Lenses map “bundles” of rays from points on the scene to the sensor.

How does this mapping work exactly?

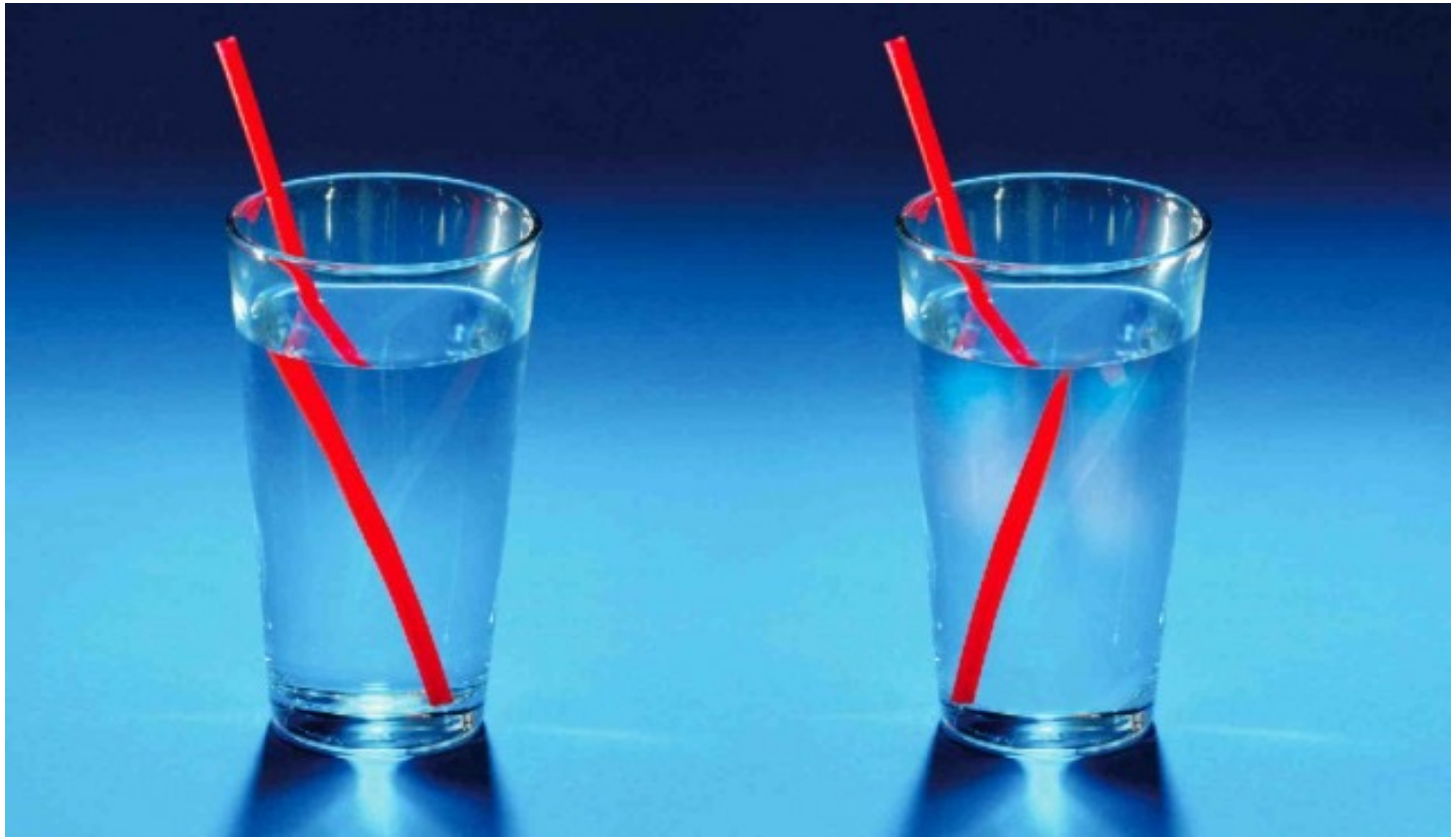
The lens on your camera

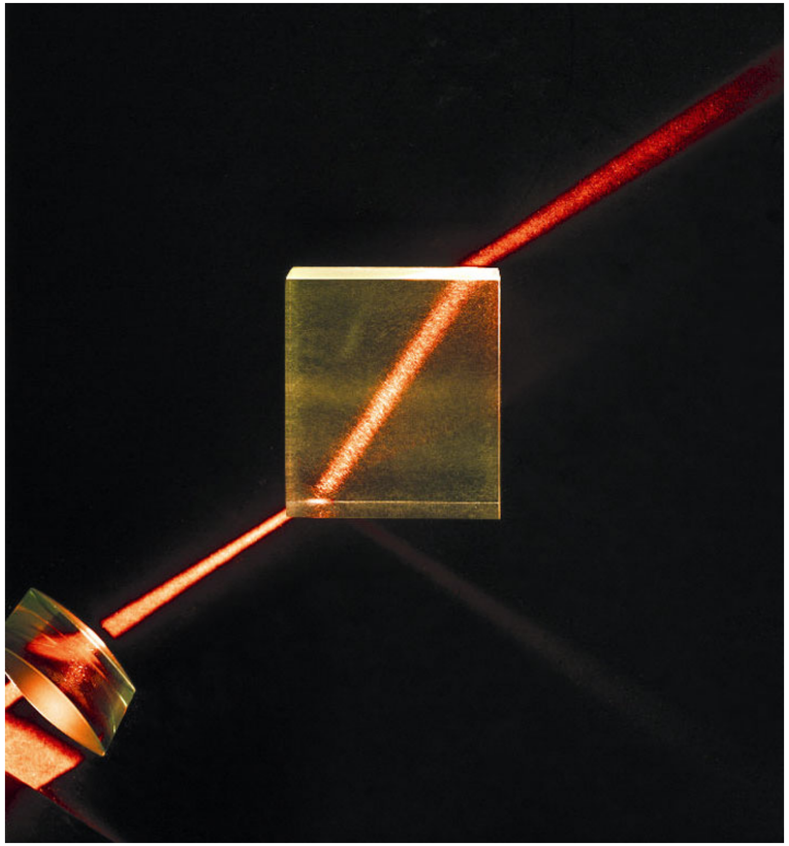
How does a lens work?



Refraction

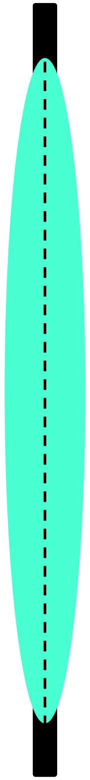
Refraction is the bending of rays of light when they move from one material to another





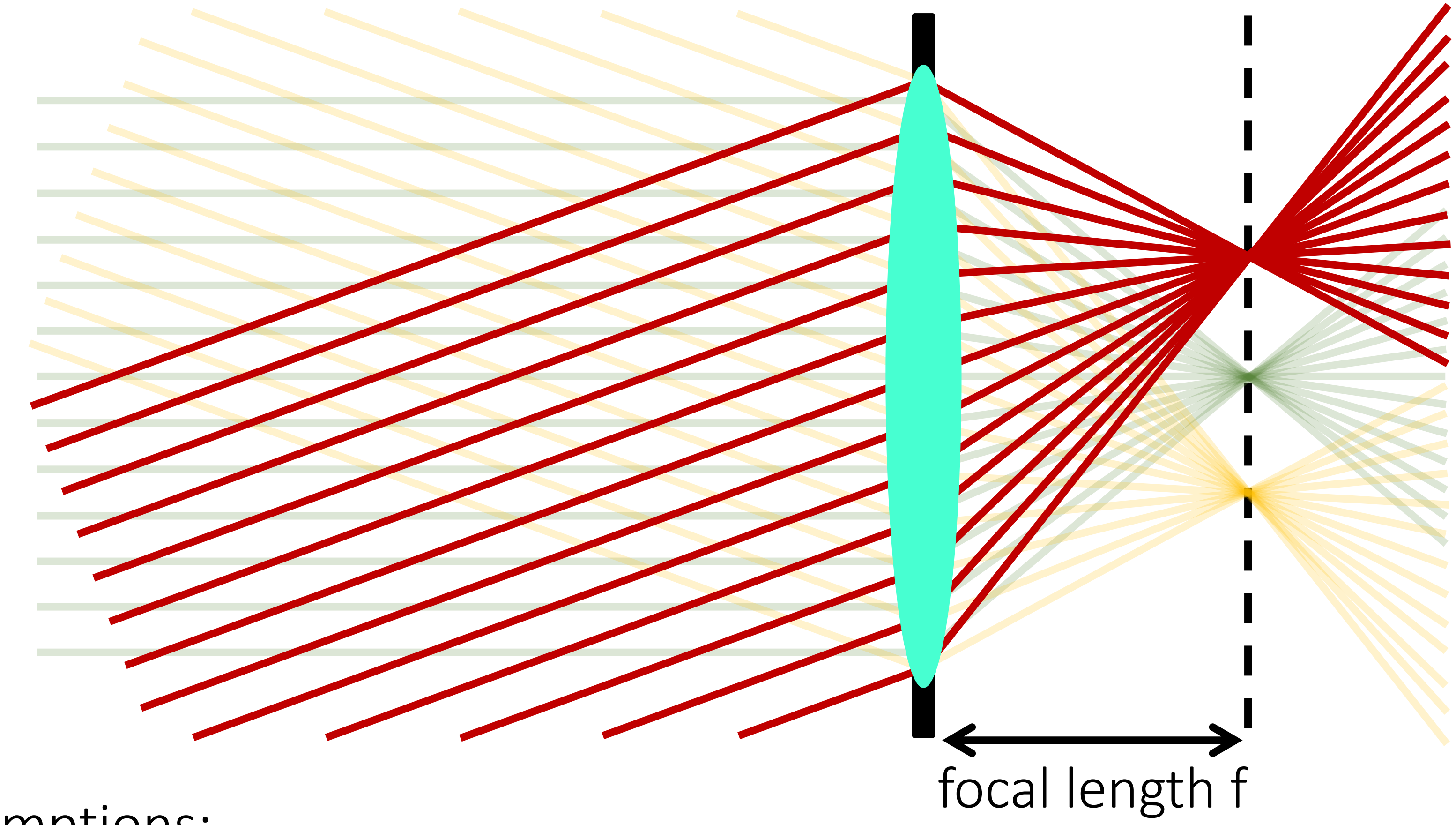
Thin lens model

Idealized geometric optics model for <u>well-designed</u> lenses.



Thin lens model

Simplification of geometric optics for well-designed lenses.

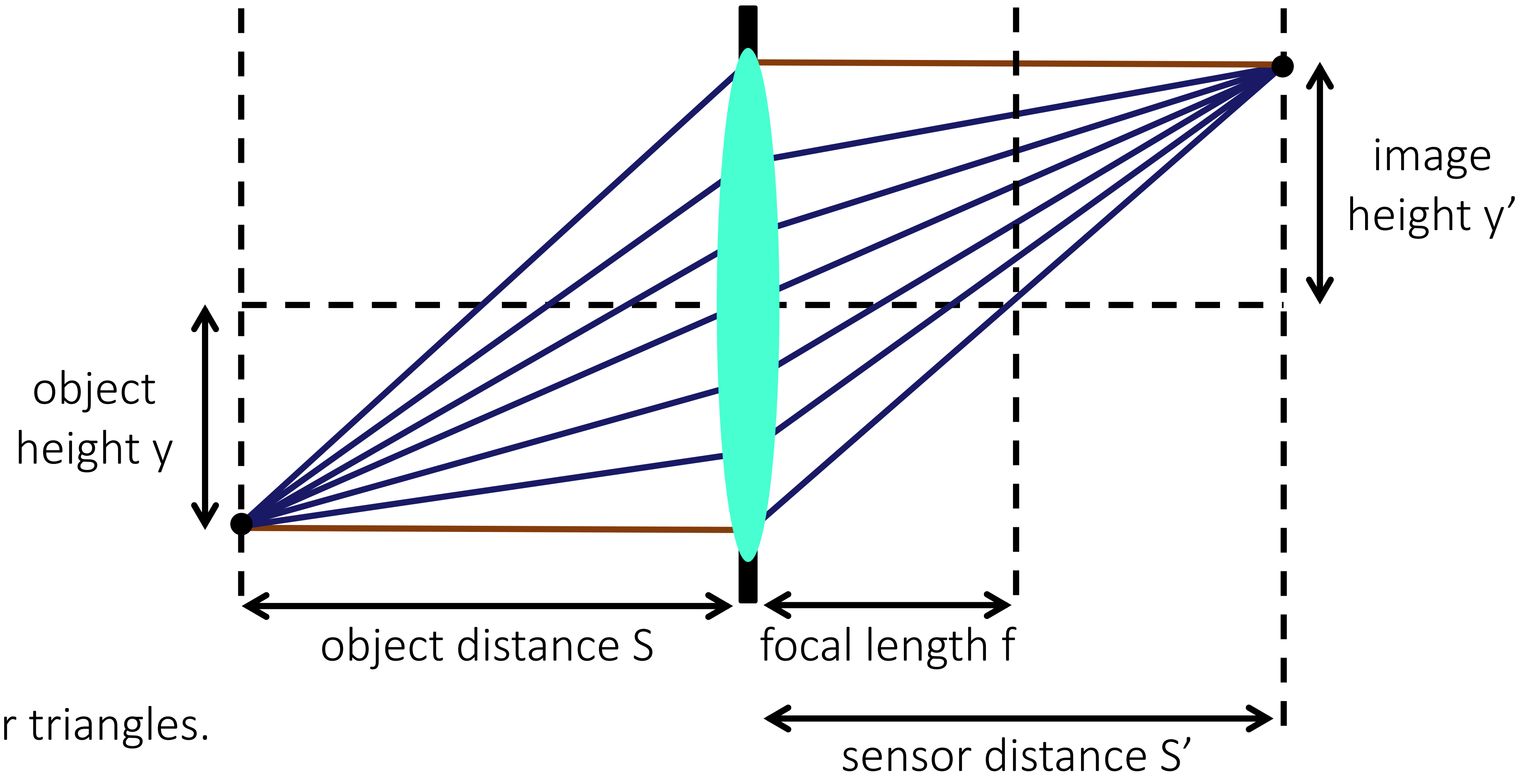


Two assumptions:

1. Rays passing through lens center are unaffected.
2. Parallel rays converge to a single point located on focal plane.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S’, y’) quantities?

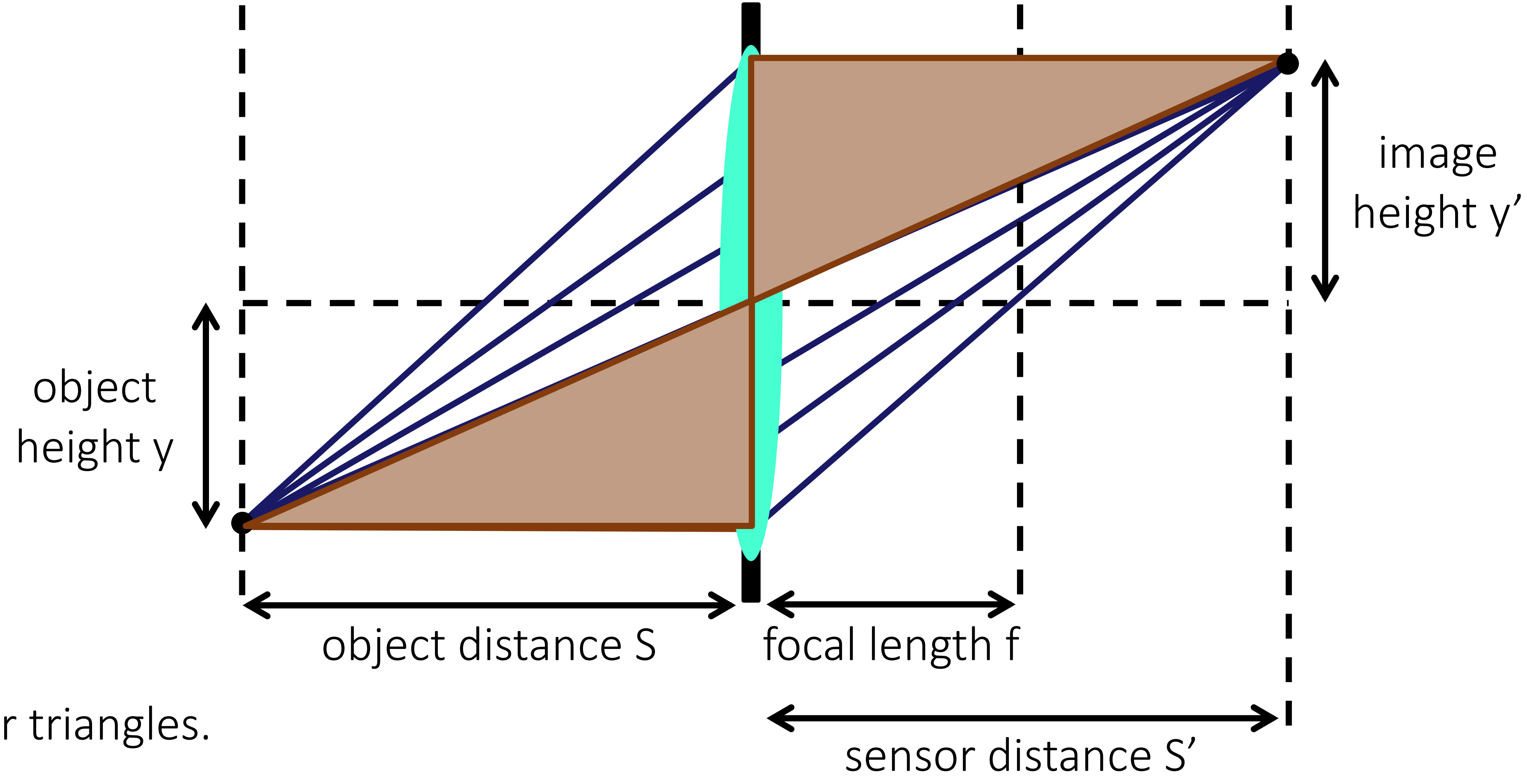


Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{y'}{y} = ?$$

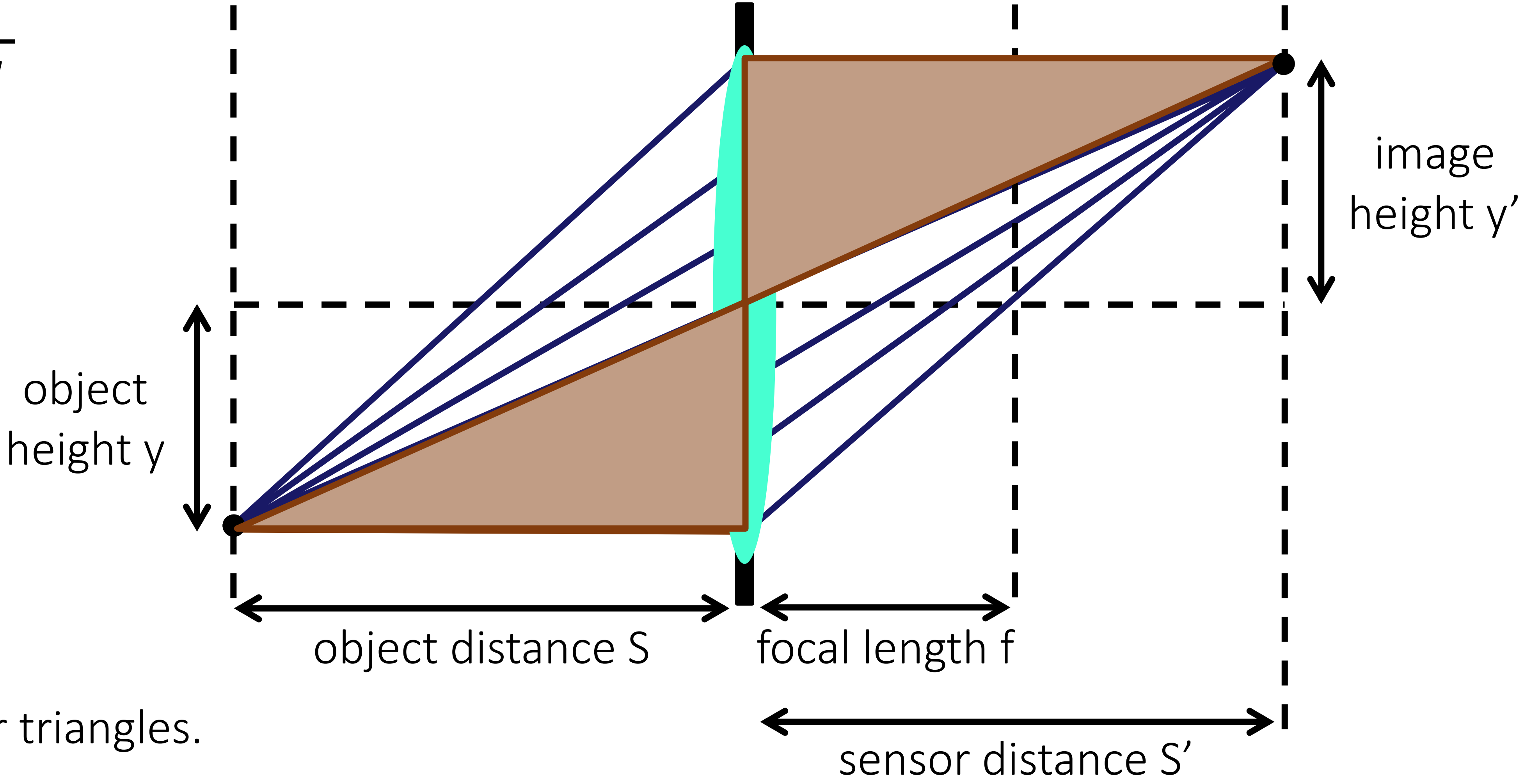


Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S’, y’) quantities?

$$\frac{y'}{y} = \frac{S'}{S}$$



Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

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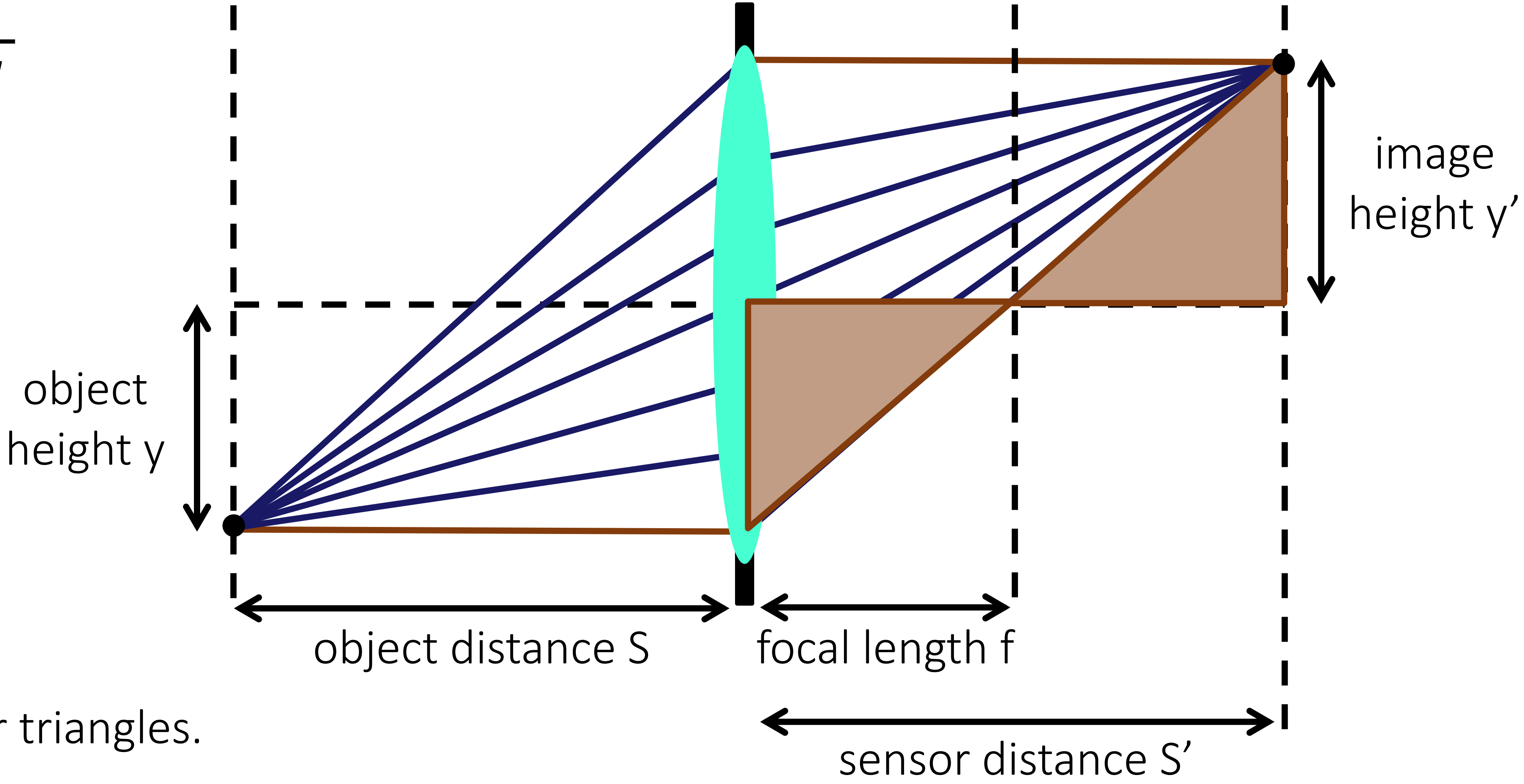


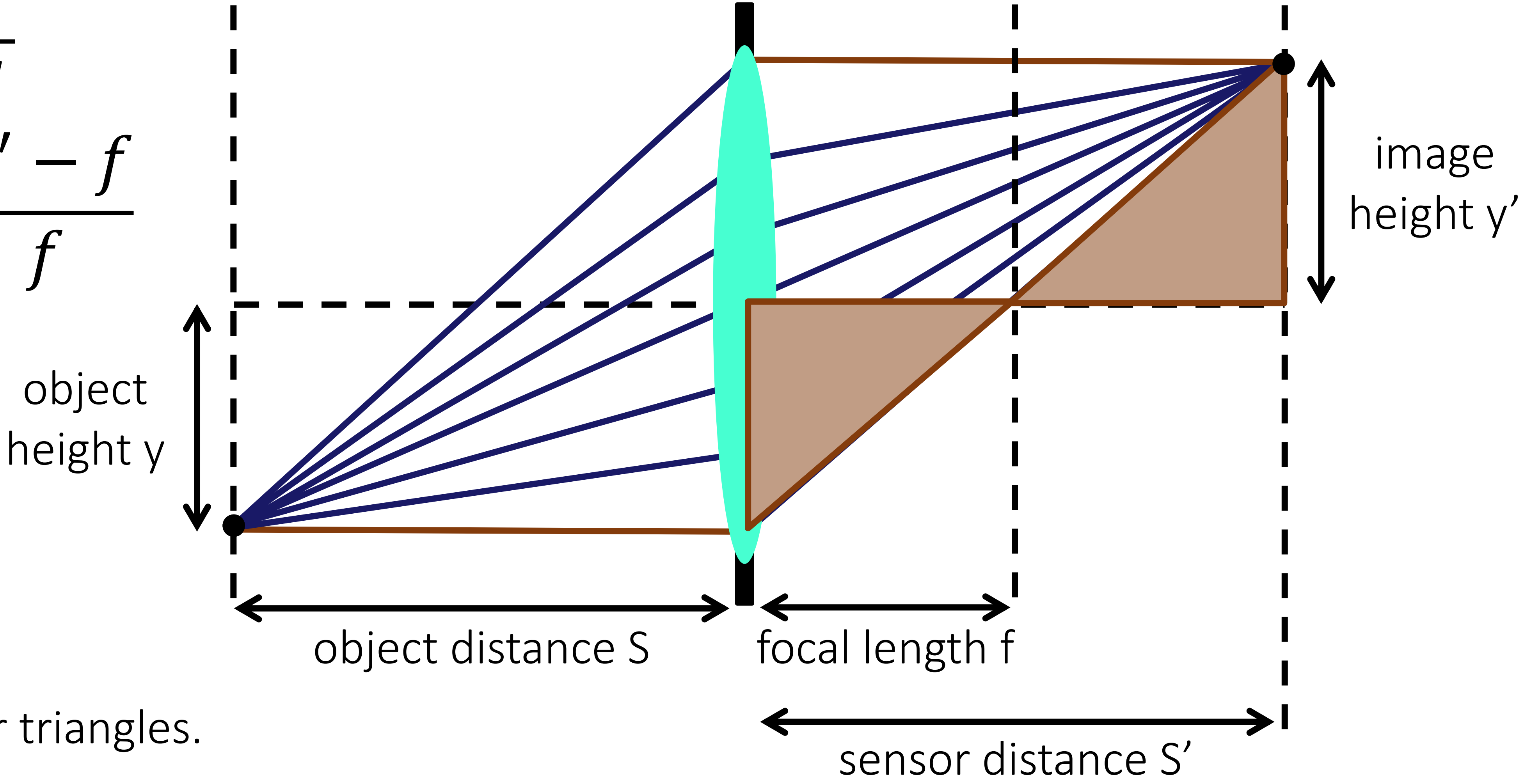
Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S’, y’) quantities?

$$\frac{y'}{y} = \frac{S'}{S}$$

$$\frac{y'}{y} = \frac{S' - f}{f}$$



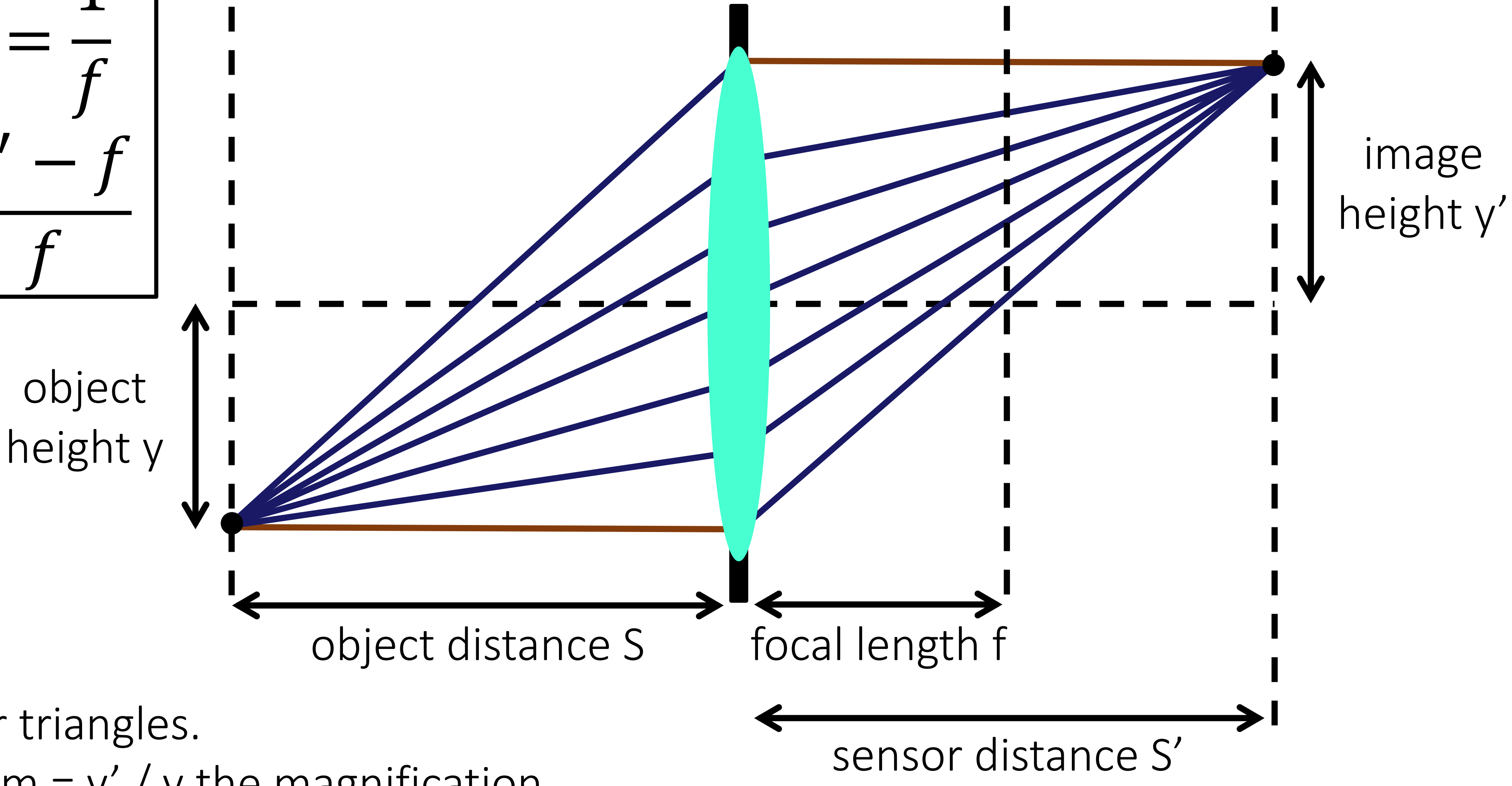
Use similar triangles.

Gaussian lens formula

How can we relate scene-space (S, y) and image space (S', y') quantities?

$$\frac{1}{S'} + \frac{1}{S} = \frac{1}{f}$$

$$m = \frac{S' - f}{f}$$



Use similar triangles.

- We call m = y' / y the magnification.

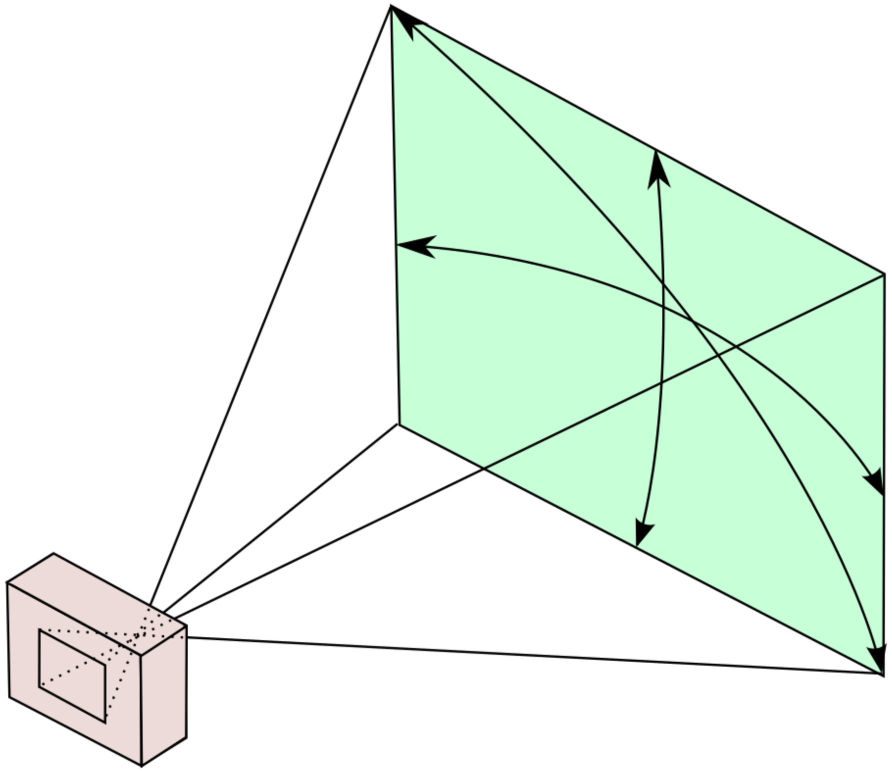
Depth of field

- Depth of field is the distance between the nearest and farthest objects in a scene that appear acceptably sharp in an image (Wikipedia)

Image source
(via A. Efros)

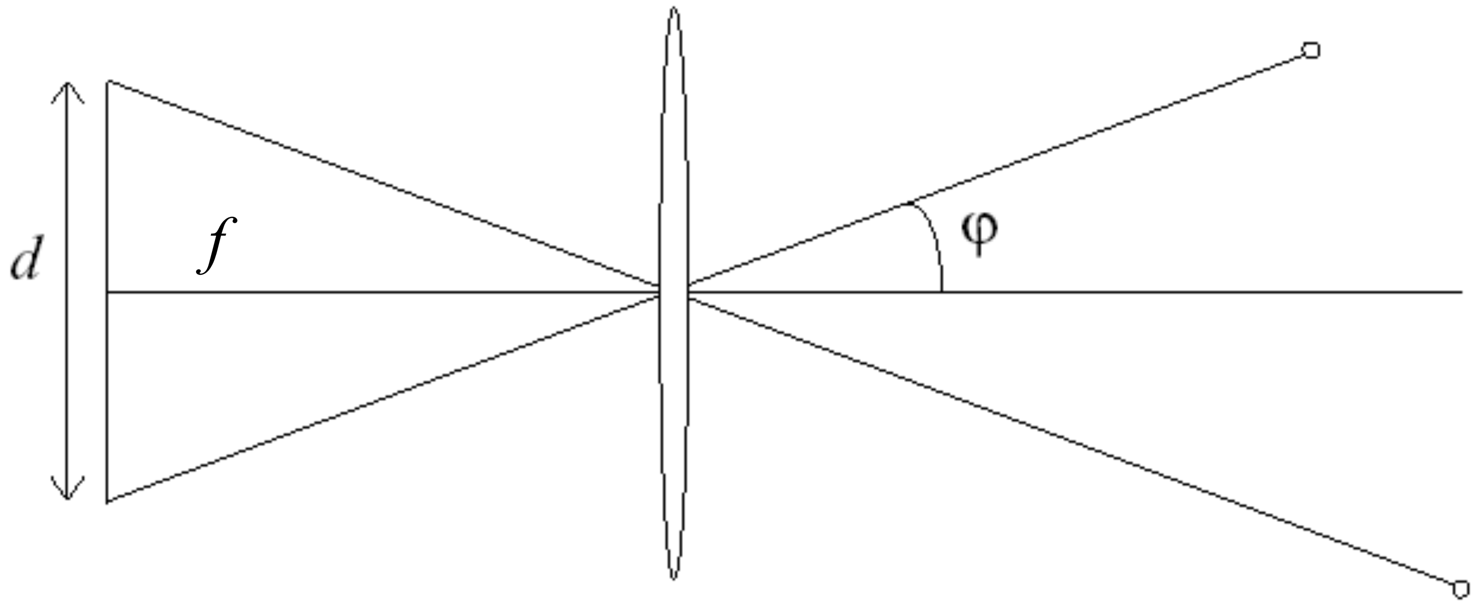
Field of view

- The field of view is the angular extent of the world observed by the camera (Wikipedia)
- What determines the FOV?



Field of view

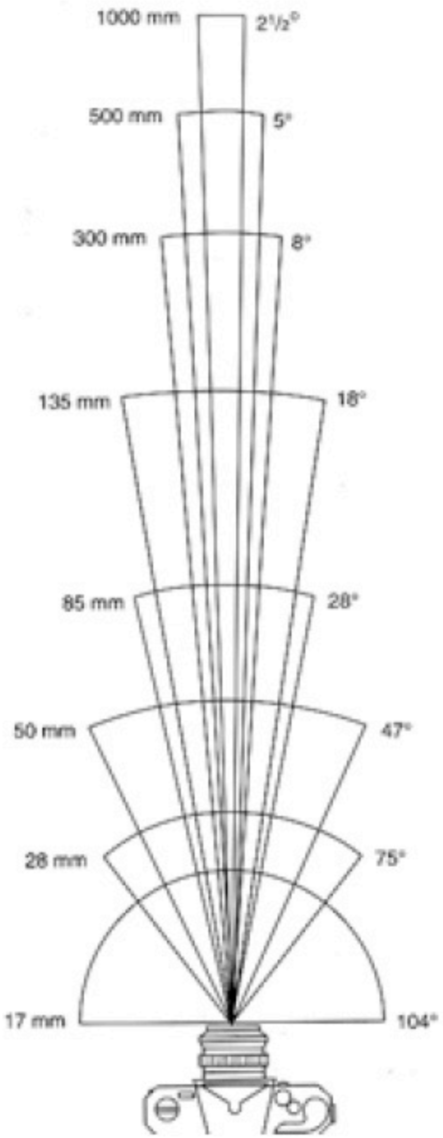
- The field of view is the angular extent of the world observed by the camera (Wikipedia)
- What determines the FOV?
  - Focal length (f), length of the sensor (d):

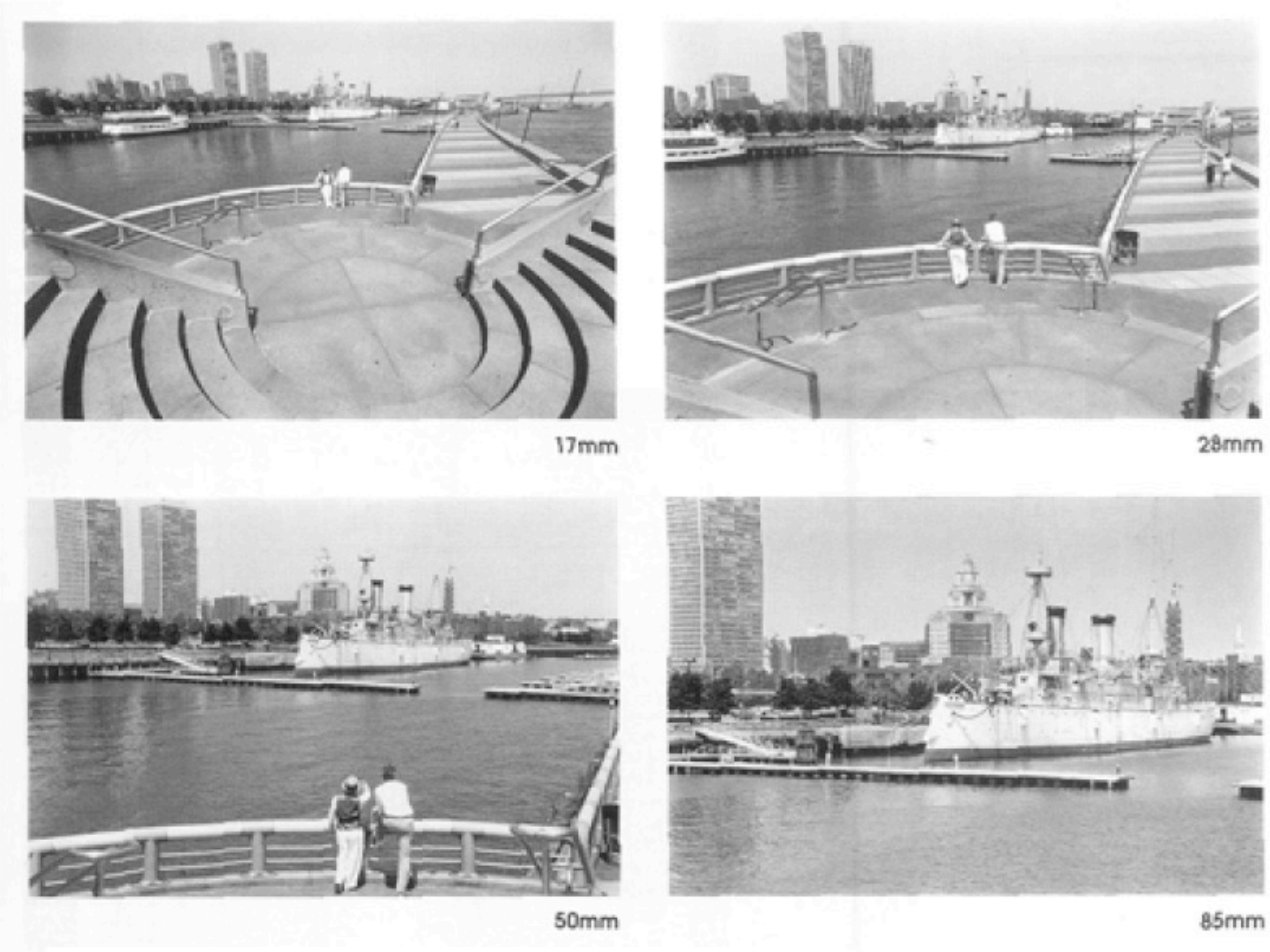


$$\varphi = \tan^{-1}\frac{d}{2f}$$

  - Larger focal length = smaller FOV

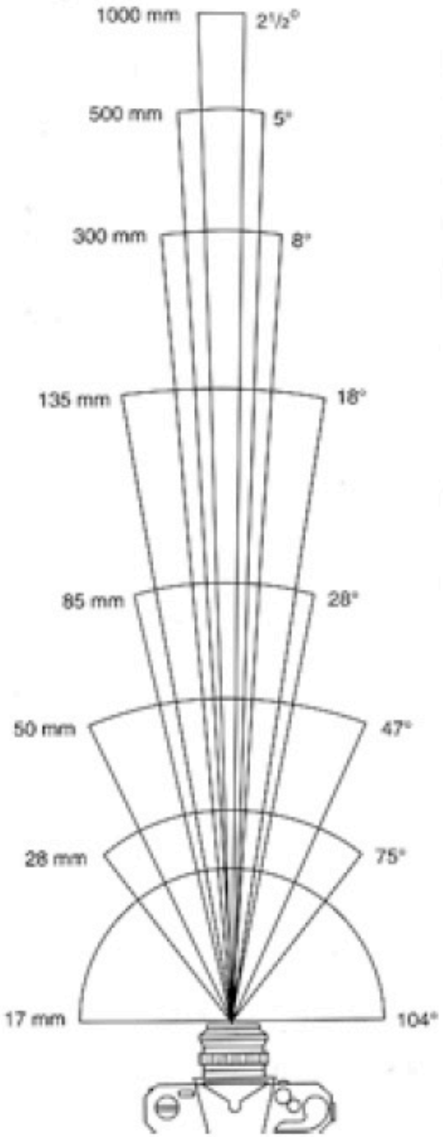
Field of view

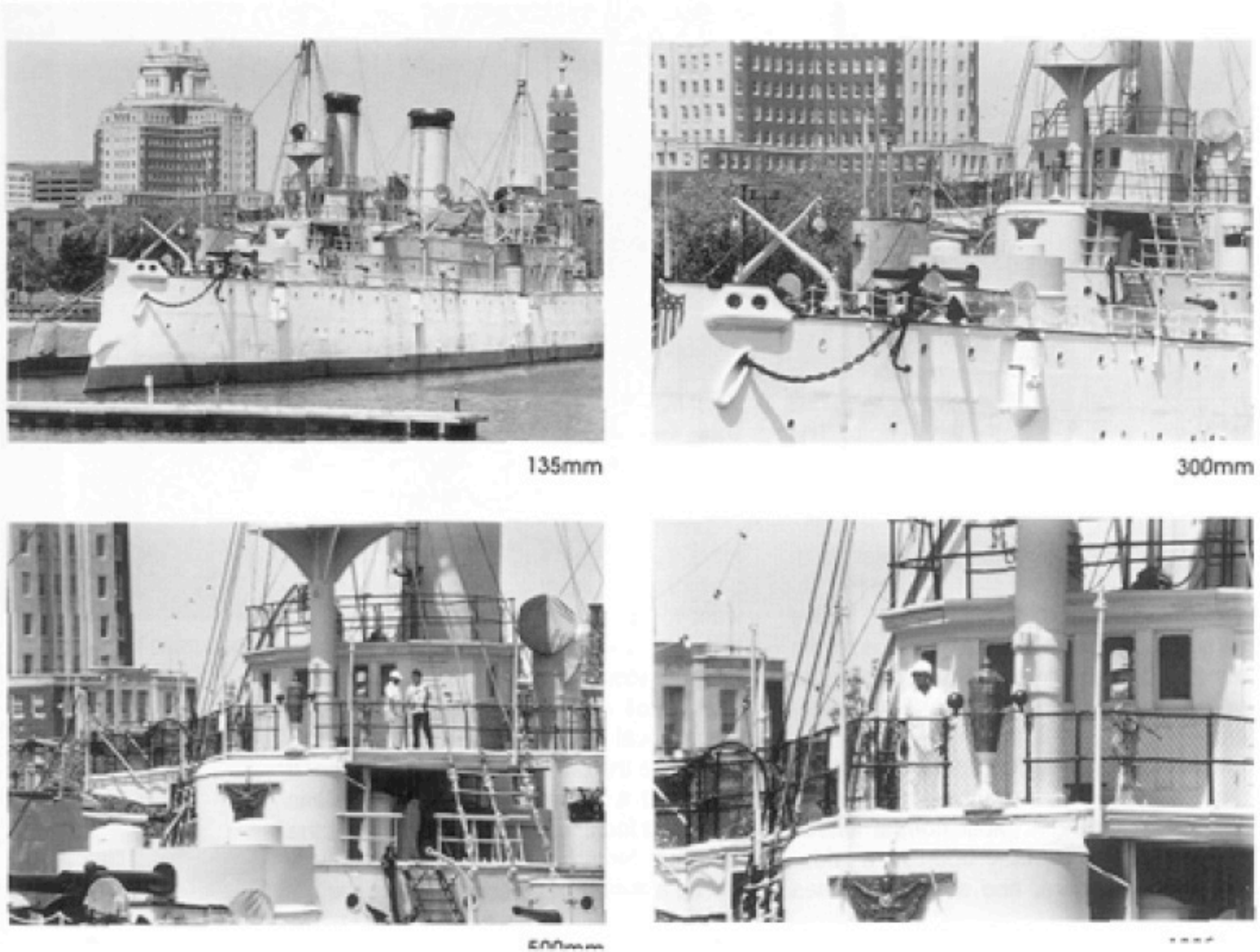




Slide by A. Efros

Field of view





Slide by A. Efros

Same effect for faces

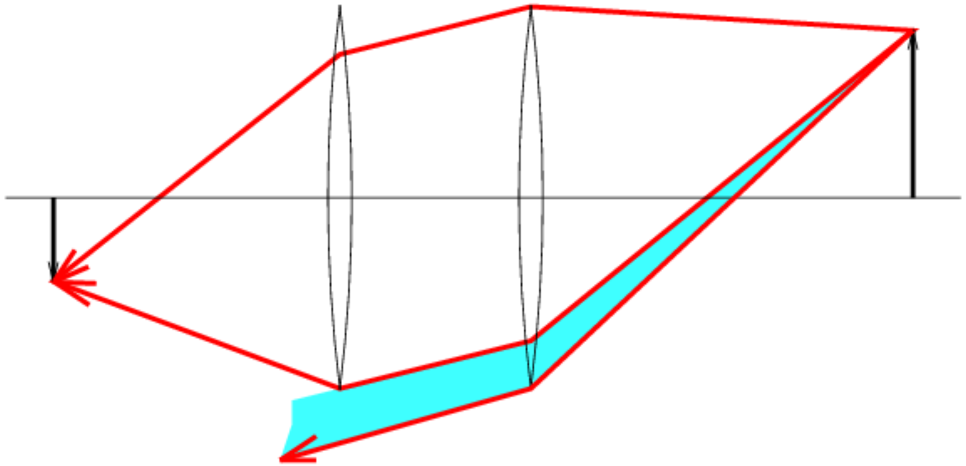
wide-angle

standard

telephoto

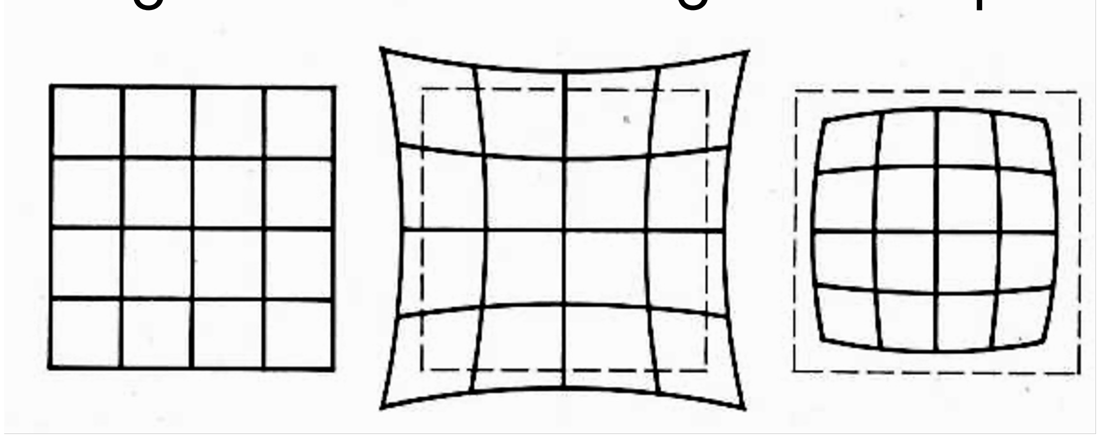
Source: F. Durand

Vignetting

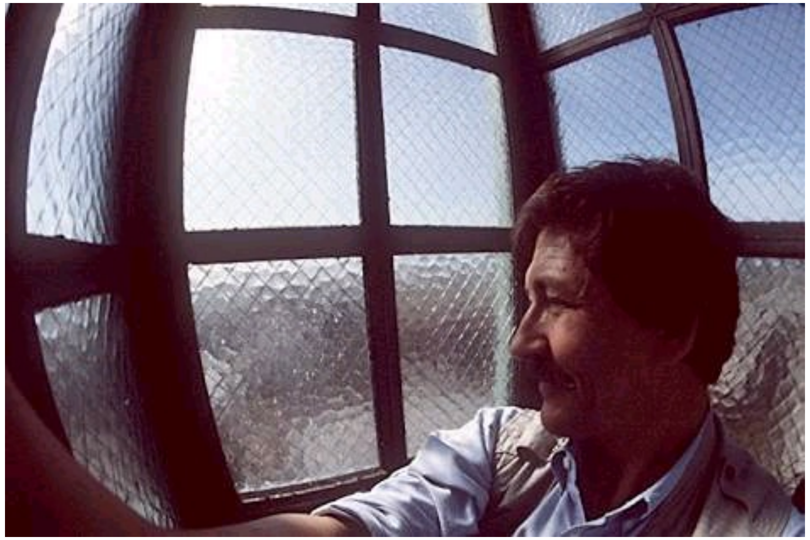


Radial distortion

- Caused by imperfect lenses
- Distortion is stronger towards the edges of the photo

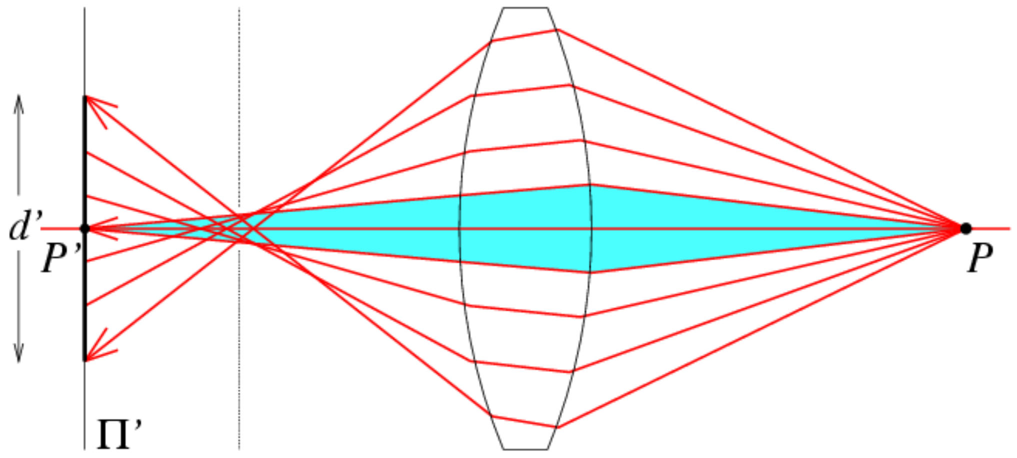


No distortion Pin cushion Barrel



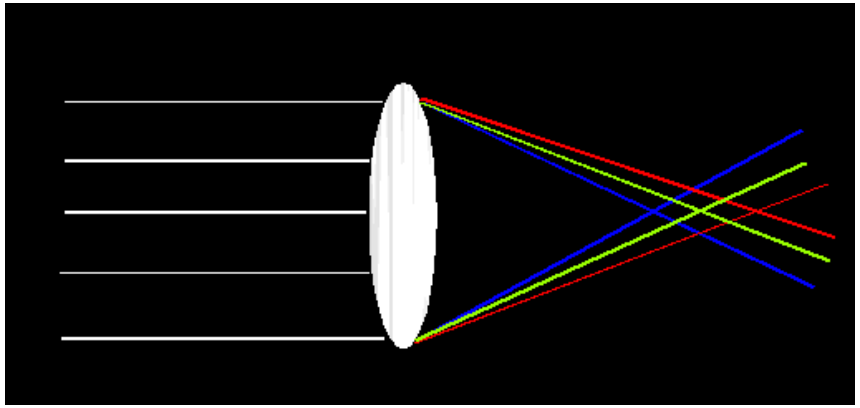
Spherical aberration

- Spherical lenses don’t focus light perfectly
- Rays farther from the optical axis focus closer



Lens flaws: Chromatic aberration

- Lens has different refractive indices for different wavelengths: causes color fringing



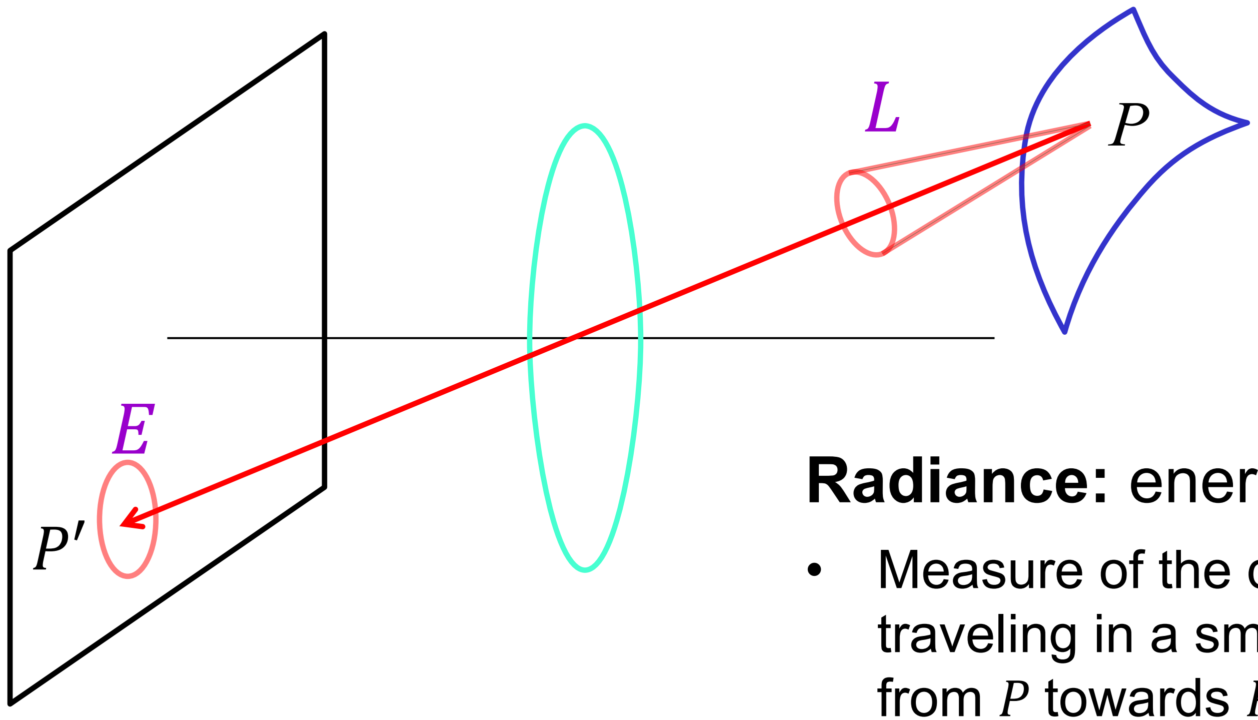
Near Lens Center

Near Lens Outer Edge

Today

- Small taste of radiometry
- Reflectance properties of surfaces
- Diffuse and specular reflection
- Shape from shading
- Estimating direction of light sources

Radiometry of image formation



Radiance: energy carried by a ray

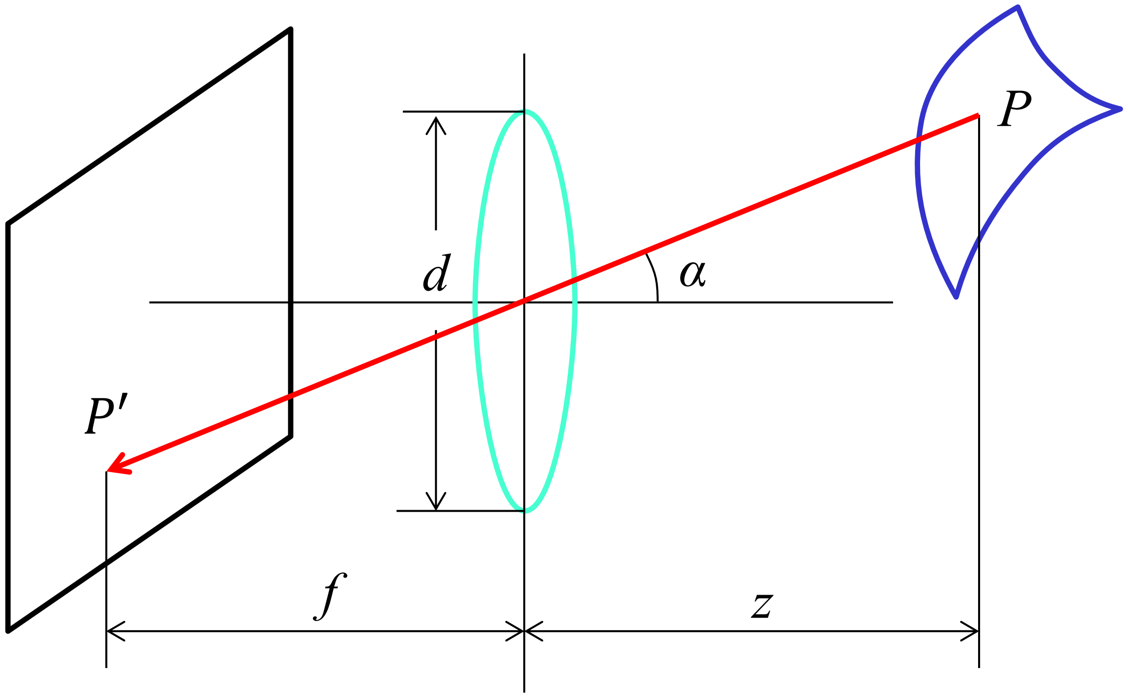
- Measure of the density of photons traveling in a small cone of directions from P towards P'
- Power per unit area perpendicular to the direction of travel, per unit solid angle
- Units: Watts per square meter per steradian

Irradiance: energy arriving at a surface

- Incident power per unit area (not foreshortened)
- Units: Watts per square meter

What is the relationship between E and L?

Fundamental radiometric relation

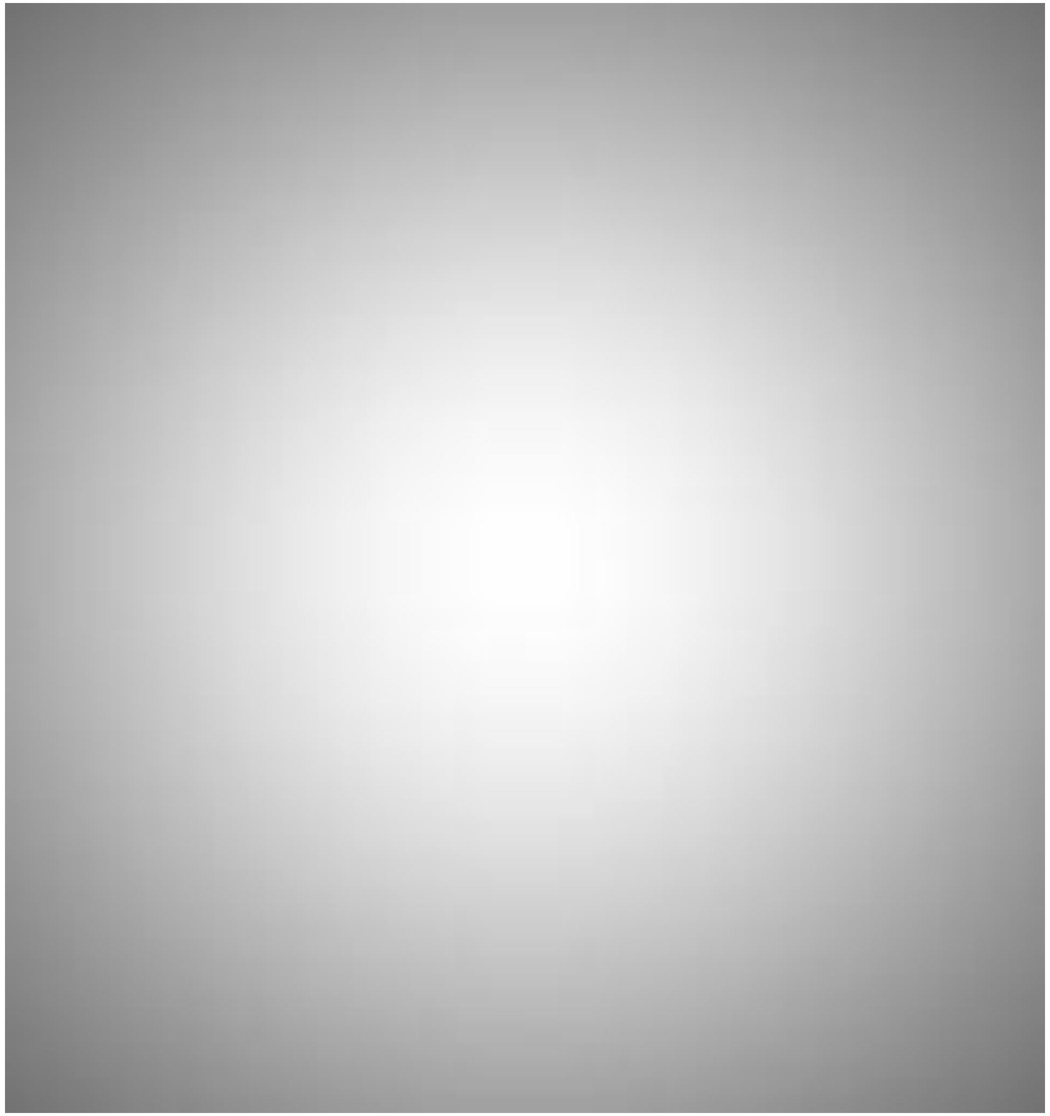


$$\boxed{E = \left[\frac{\pi}{4}\left(\frac{d}{f}\right)^2 \cos^4\alpha\right] L}$$

- Image irradiance (E) is linearly related to scene radiance (L)
- Irradiance is *directly* proportional to the area of the lens ($\frac{\pi d^2}{4}$) and *inversely* proportional to the squared distance between the lens and the image plane (f)
- The irradiance decreases as the angle between the viewing ray and the optical axis (α) increases

For derivation, see, e.g., Szeliski 2.2.3

Fundamental radiometric relation

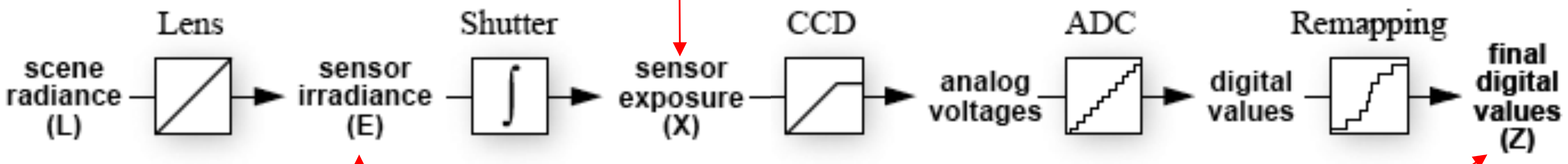


$$\boxed{E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L}$$

S. B. Kang and R. Weiss. [Can we calibrate a camera using an image of a flat, textureless Lambertian surface?](#) ECCV 2000

From light rays to pixel values

$$X = E \cdot \Delta t$$



$$E = \left[\frac{\pi}{4}\left(\frac{d}{f}\right)^2 \cos^4\alpha\right] L$$

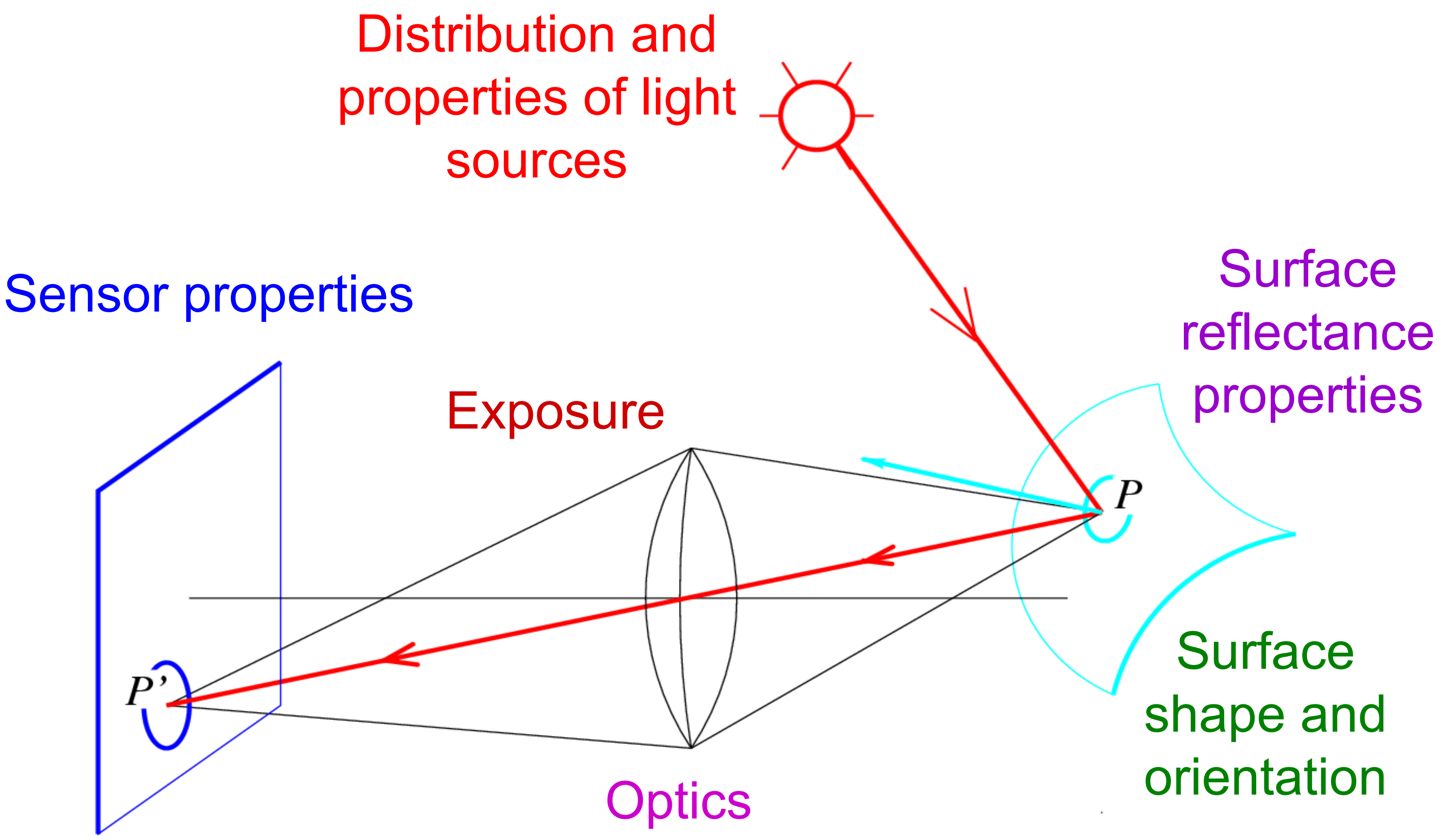
$$Z = f(E \cdot \Delta t)$$

- **Camera response function**: the mapping f from irradiance to pixel values
  - Needed for applications like estimation of scene reflectance properties, creating high dynamic range (HDR) images
  - For further reading: M. Brown, Understanding the In-Camera Image Processing Pipeline for Computer Vision, CVPR 2016 Tutorial

Figure source: P. Debevec and J. Malik. Recovering High Dynamic Range Radiance Maps from Photographs. SIGGRAPH 1997

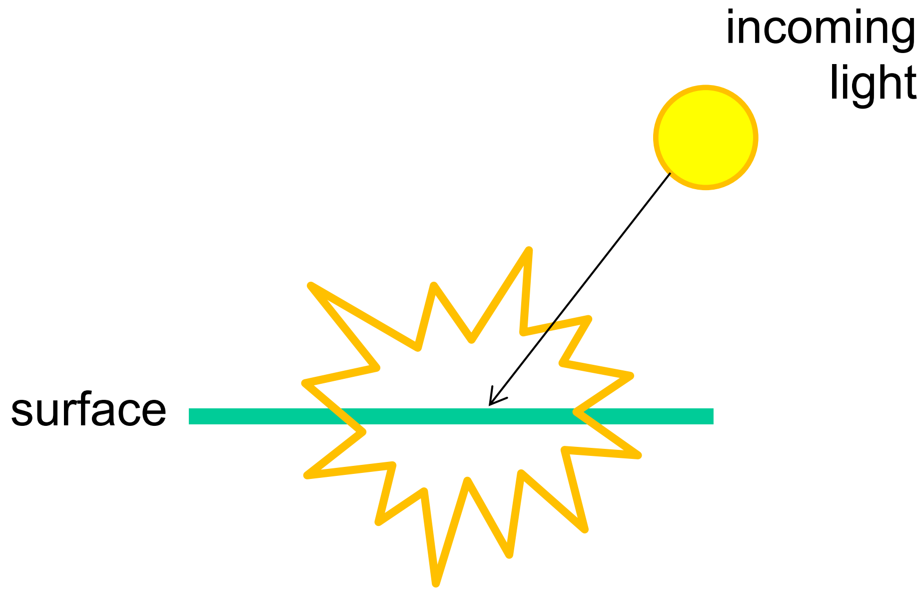
Recall: Image formation

- What determines the brightness of an image pixel?



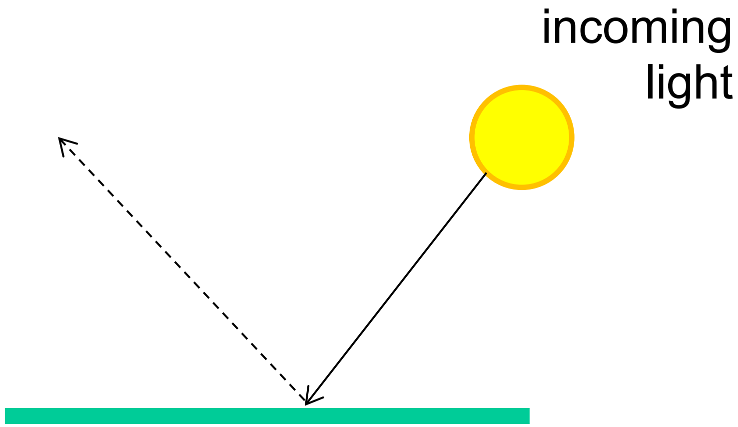
Slide by L. Fei-Fei

What can happen to light when it hits a surface?

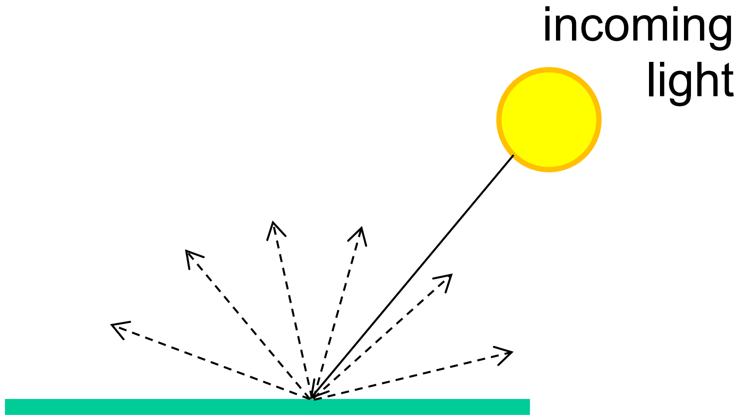


Basic models of reflection

- **Specular reflection:** light is reflected about the surface normal



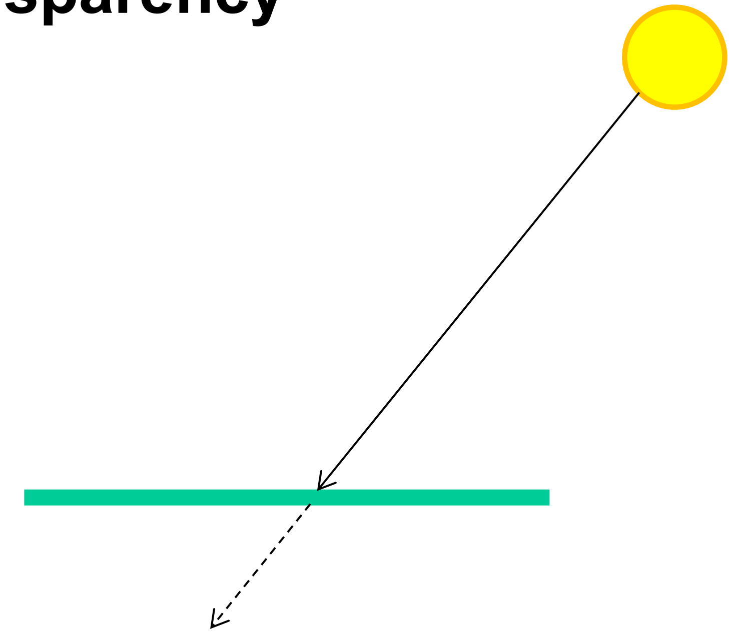
- **Diffuse reflection:** light scatters equally in all directions



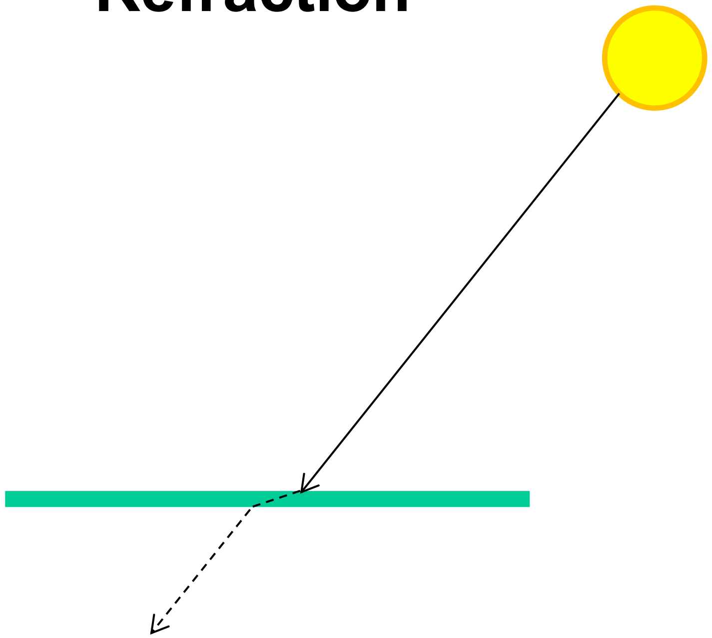
Slide from D. Hoiem

Other possible effects

- **Transparency**



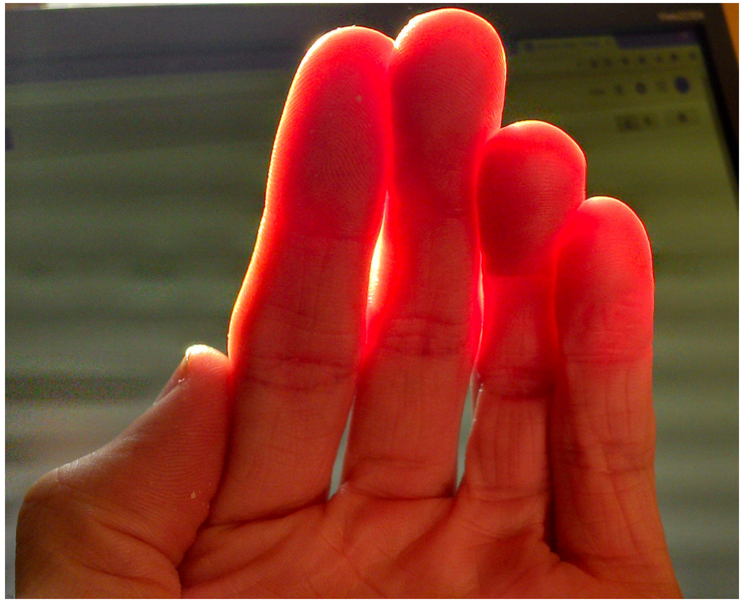
- **Refraction**

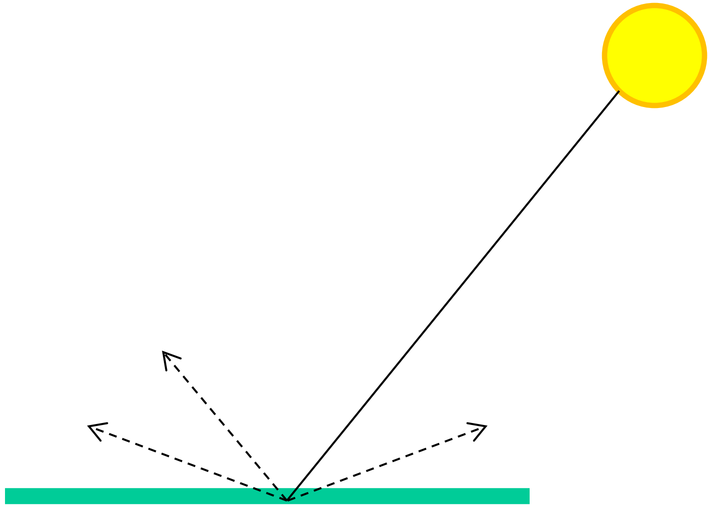


Slide from D. Hoiem

Other possible effects

- **Subsurface scattering**



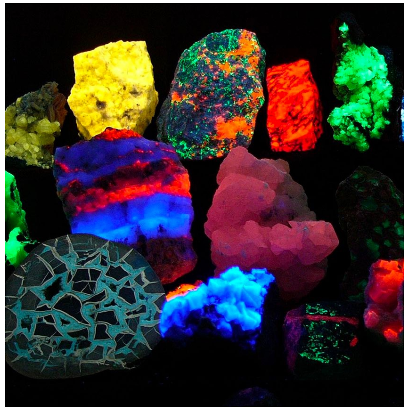


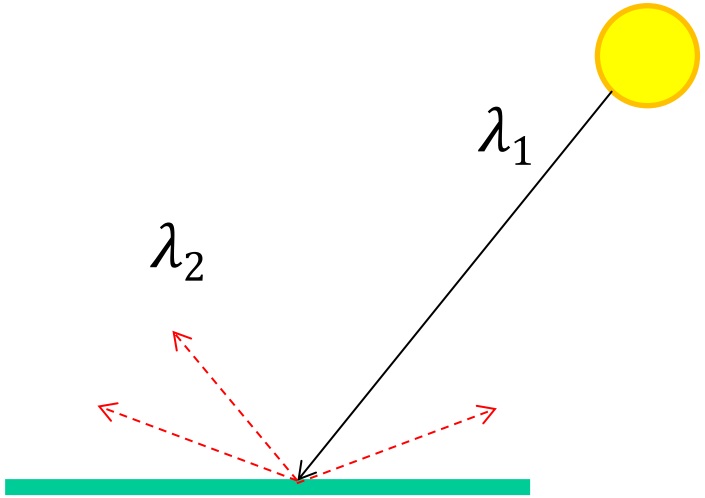
Slide from D. Hoiem

Image source

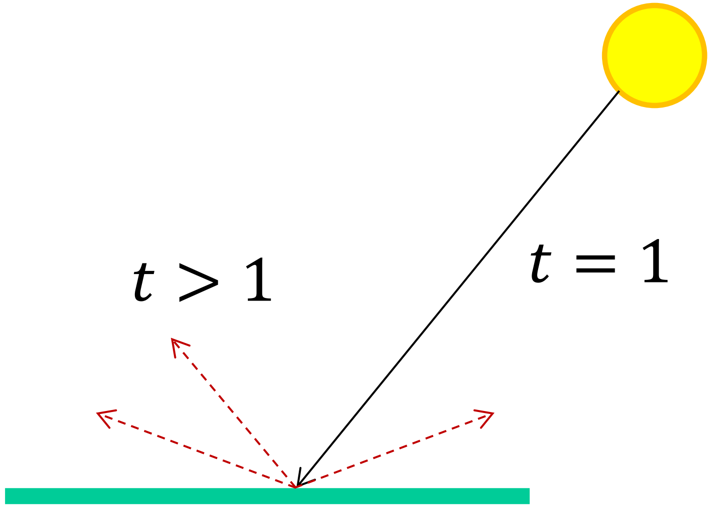
Other possible effects

- **Fluorescence**





- **Phosphorescence**

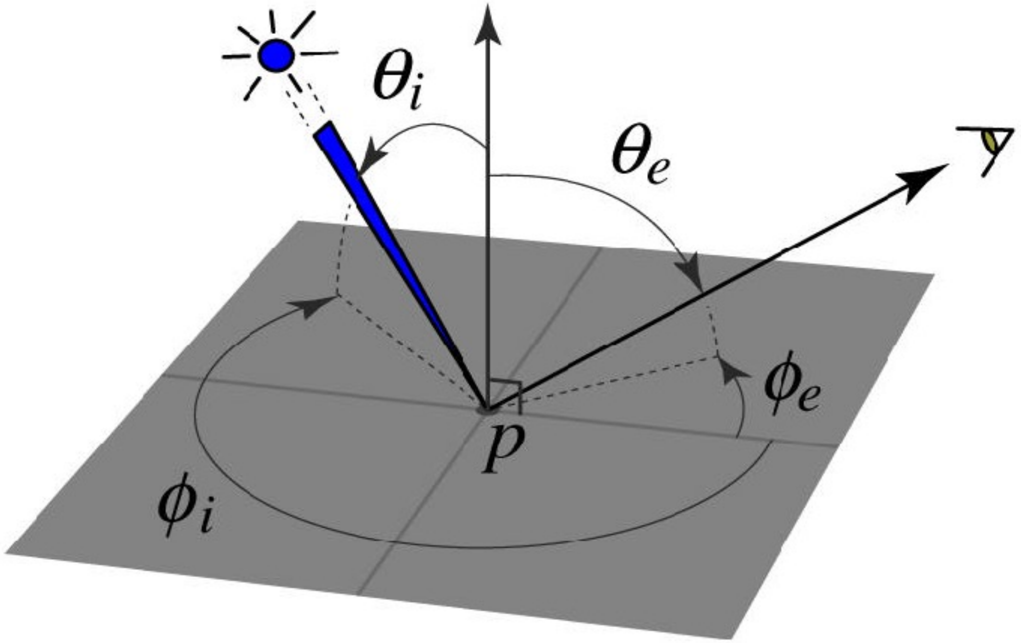


Slide from D. Hoiem

Image source

Bidirectional reflectance distribution function (BRDF)

- How bright a surface appears when viewed from one direction when light falls on it from another
- Definition: ratio of the radiance in the emitted direction to irradiance in the incident direction



Function of (at least) four parameters: incident and outgoing θ, ϕ

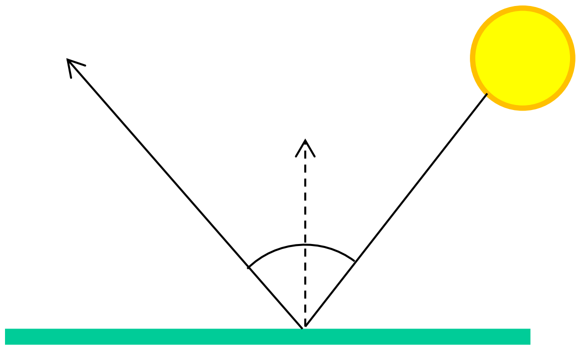
Source: Steve Seitz

Bidirectional reflectance distribution function (BRDF)

- How bright a surface appears when viewed from one direction when light falls on it from another
- Definition: ratio of the radiance in the emitted direction to irradiance in the incident direction
- Can be incredibly complicated!

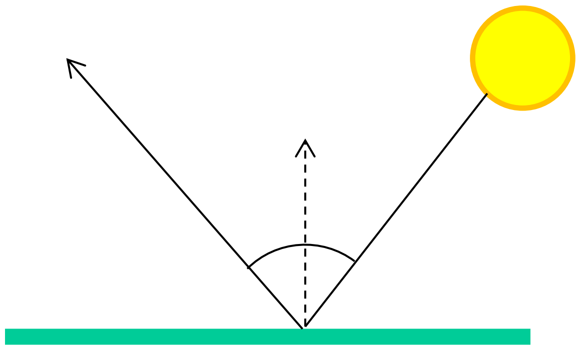
Specular reflection

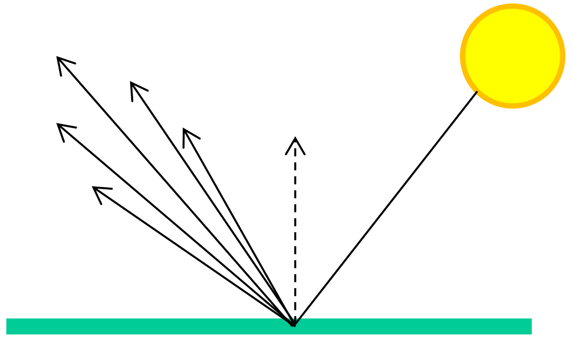
- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)



Specular reflection

- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)
- On real surfaces, energy usually goes into a "lobe" of directions





Specular reflection

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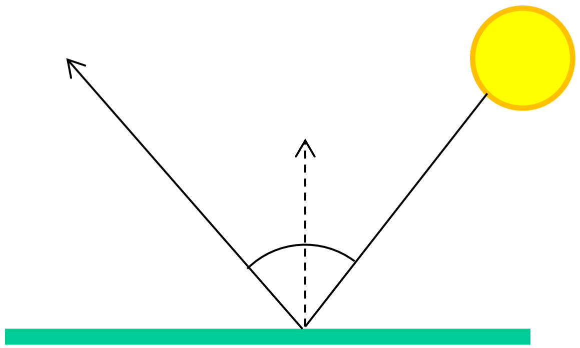
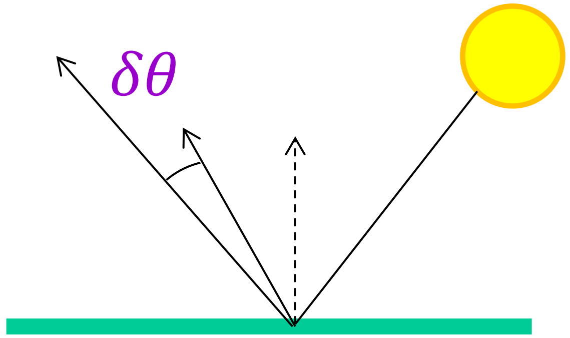
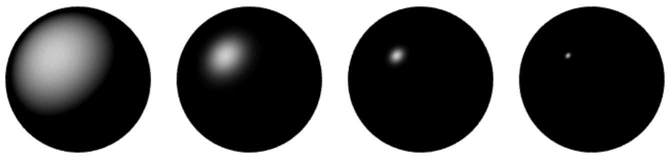
- On real surfaces, energy usually goes into a "lobe" of directions
- **Phong model:** reflected energy falls of with $\cos^n(\delta\theta)$

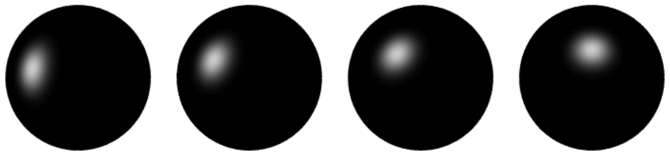




Changing the exponent

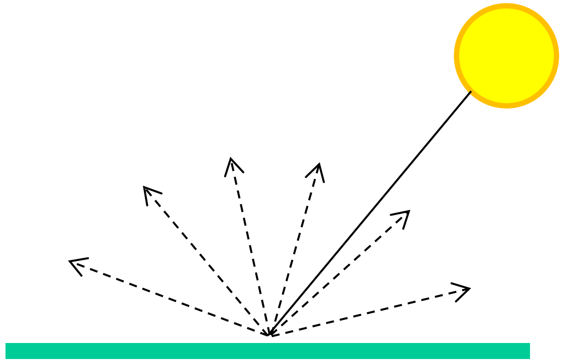


Moving the light source



Diffuse reflectance

- Light scatters equally in all directions
  - E.g., brick, matte plastic, rough wood



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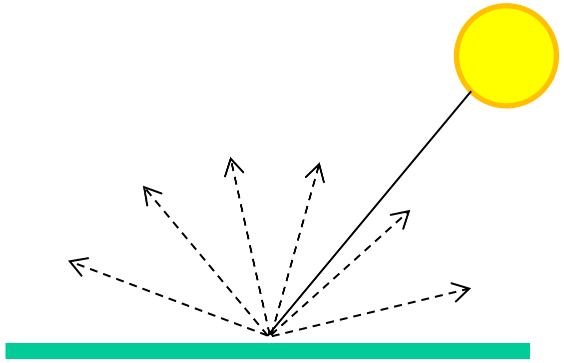


- This happens because of *microfacets* that scatter incoming light randomly

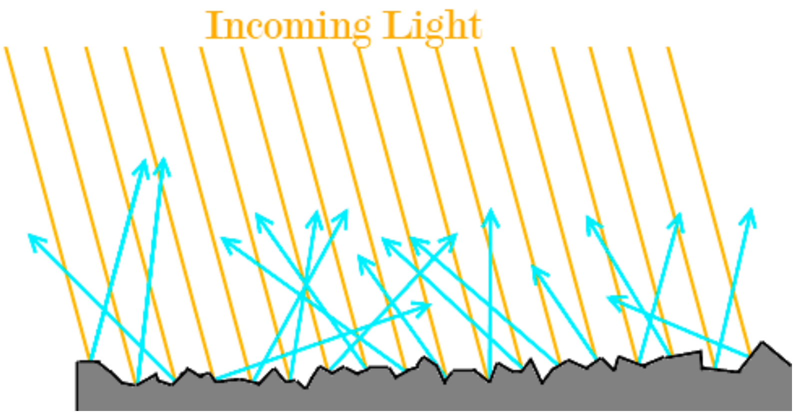
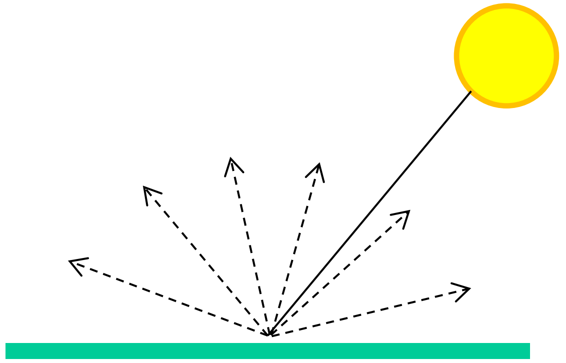


Image source

Diffuse reflectance

- Light scatters equally in all directions
  - For a fixed incidence angle, BRDF is constant



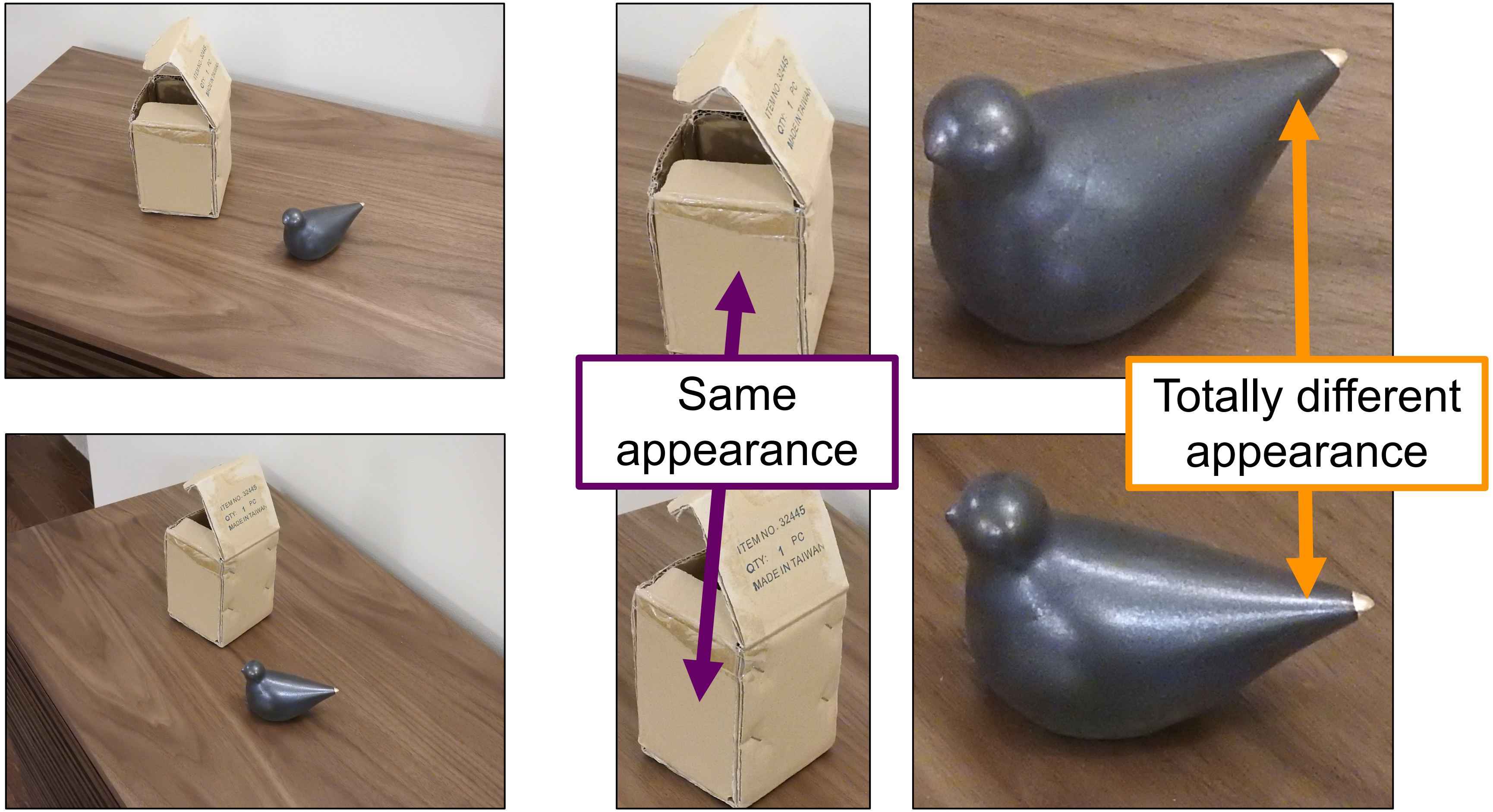
- What if we change the incidence angle?

Why do we care about diffuse reflectance?

Same lighting, as close as possible camera settings, but different **camera position**

Diffuse

Specular



Source: [J. Johnson and D. Fouhey]

Diffuse reflectance

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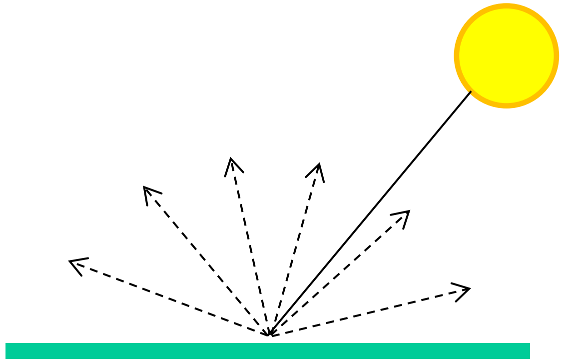
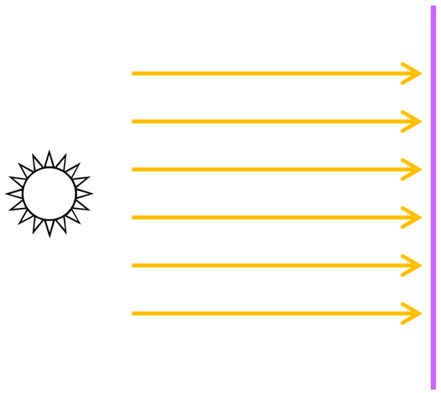
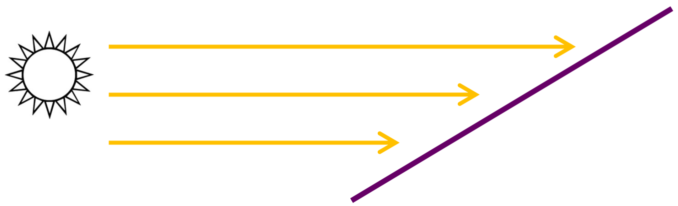


- What if we change the incidence angle?

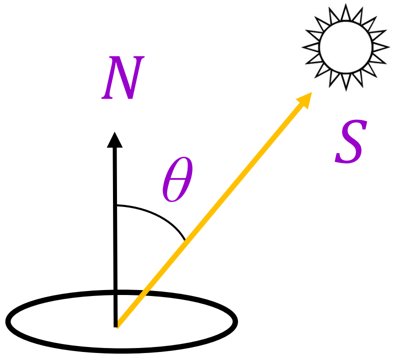


brighter

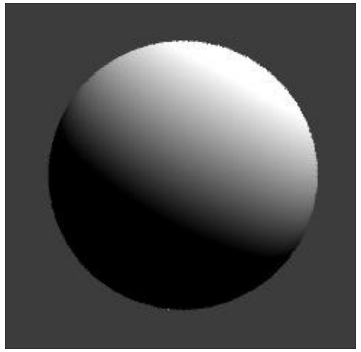


darker

Diffuse reflectance: Lambert's law



$$
\begin{aligned}
I &= \rho\,(S \cdot N) \\
&= \rho \|S\| \cos\theta
\end{aligned}
$$



I: reflected intensity (technically: *radiosity*, or total power leaving the surface per unit area)

ρ: albedo (fraction of incident irradiance reflected by the surface)

S: direction of light source (magnitude proportional to intensity of the source)

N: unit surface normal

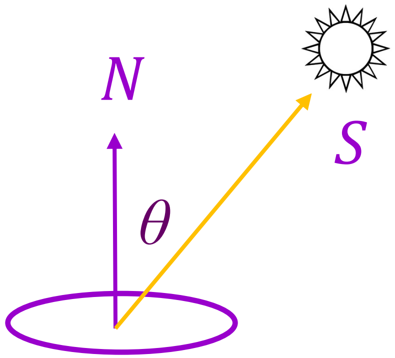
Photometric stereo, or shape from shading

- Can we reconstruct the shape of an object based on shading cues?

Luca della Robbia,
Cantoria, 1438

Photometric stereo, or shape from shading

- Can we reconstruct the shape of an object based on shading cues?
- Assuming a Lambertian object, given the image intensity (I), can we recover the light source direction (S) and the surface normal (N)?
- Can we do this from a single image?



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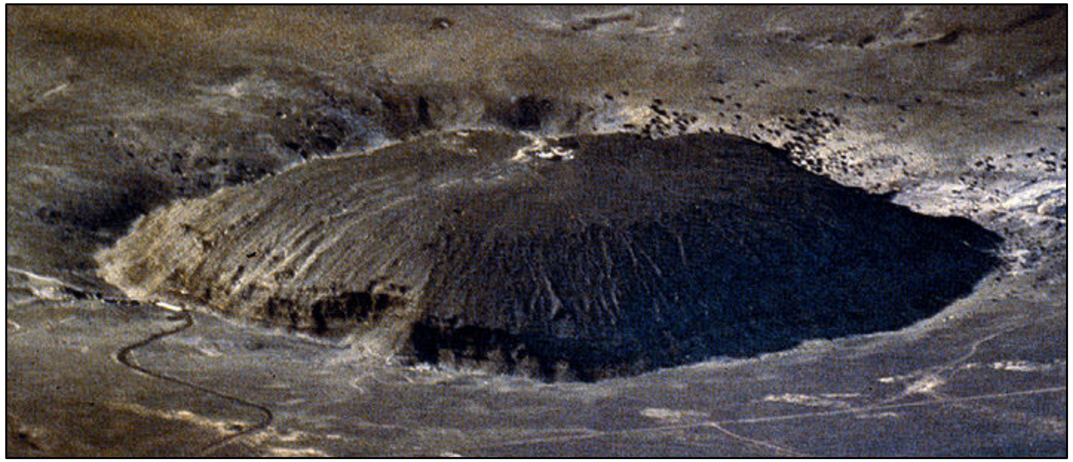
$$

Shape from shading ambiguity



Source: J. Johnson and D. Fouhey

Image source

Shape from shading ambiguity

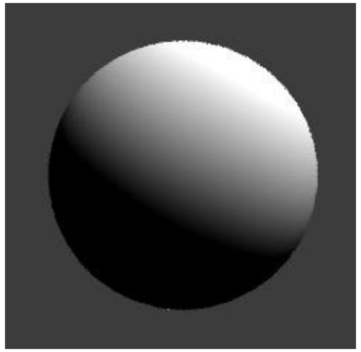
- Humans assume light from above (and the blueness also tells you distance)

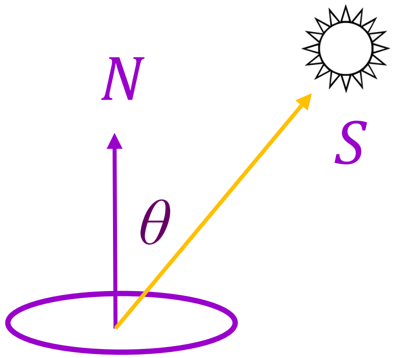
Source: [J. Johnson and D. Fouhey]

[Image source]

Review: Lambert's law

- What determines the observed brightness at a point on the surface?

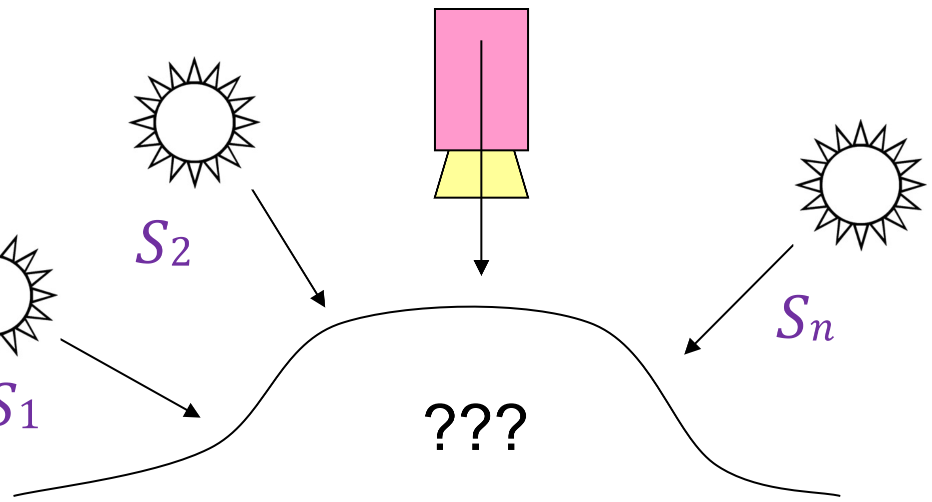




$$I = \rho\,(S \cdot N)$$
$$= \rho \|S\| \cos\theta$$

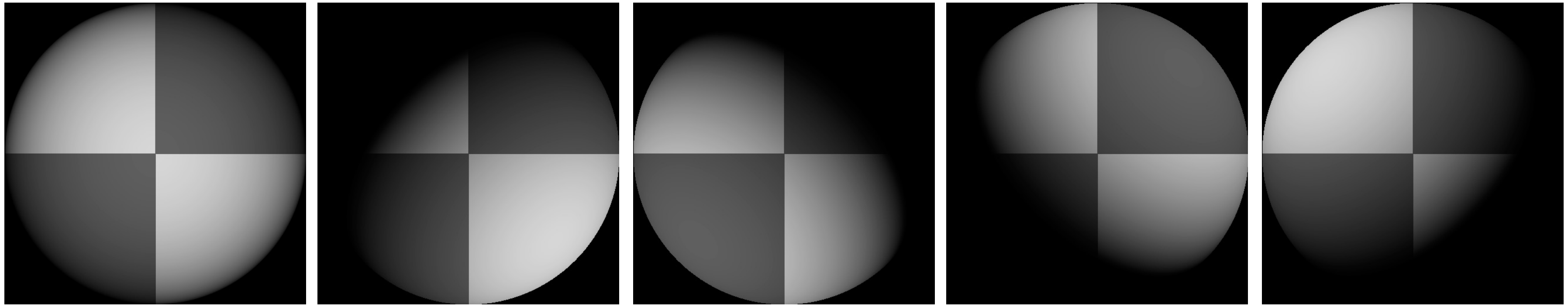
Photometric stereo

- Assume:
  - A Lambertian object
  - A *local shading model* (each point on a surface receives light only from sources visible at that point)
  - A set of *known* light source directions
  - A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources
  - Orthographic projection
- Goal: reconstruct object shape and albedo

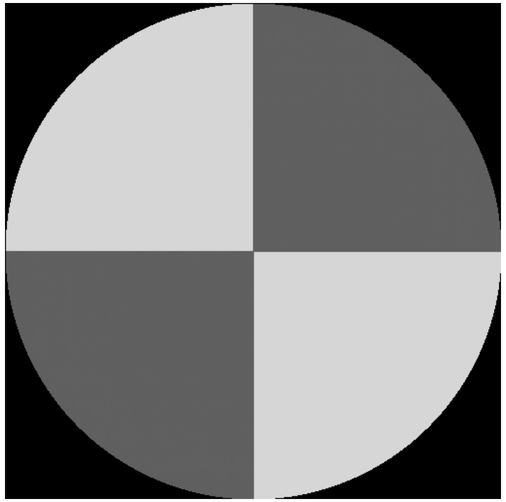


F&P 2nd ed., sec. 2.2.4

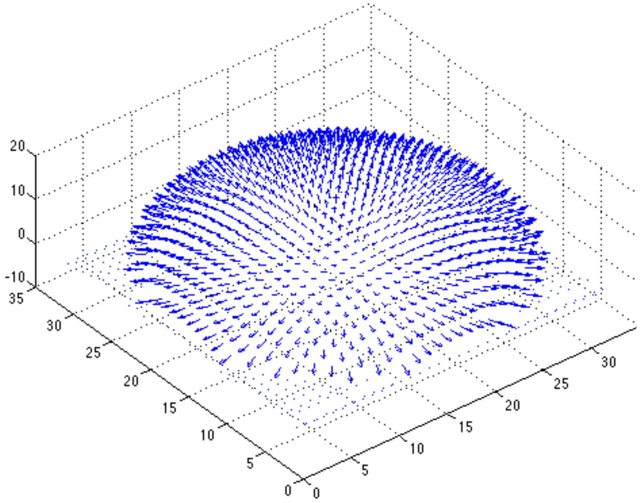
Synthetic example



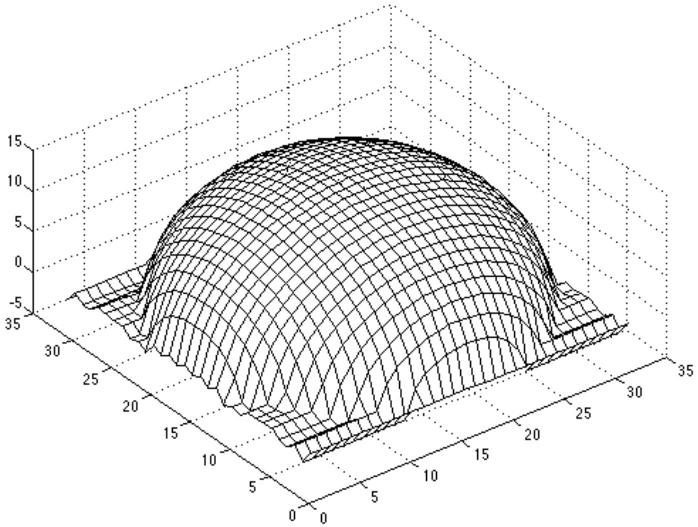
Recovered albedo



Recovered normal field



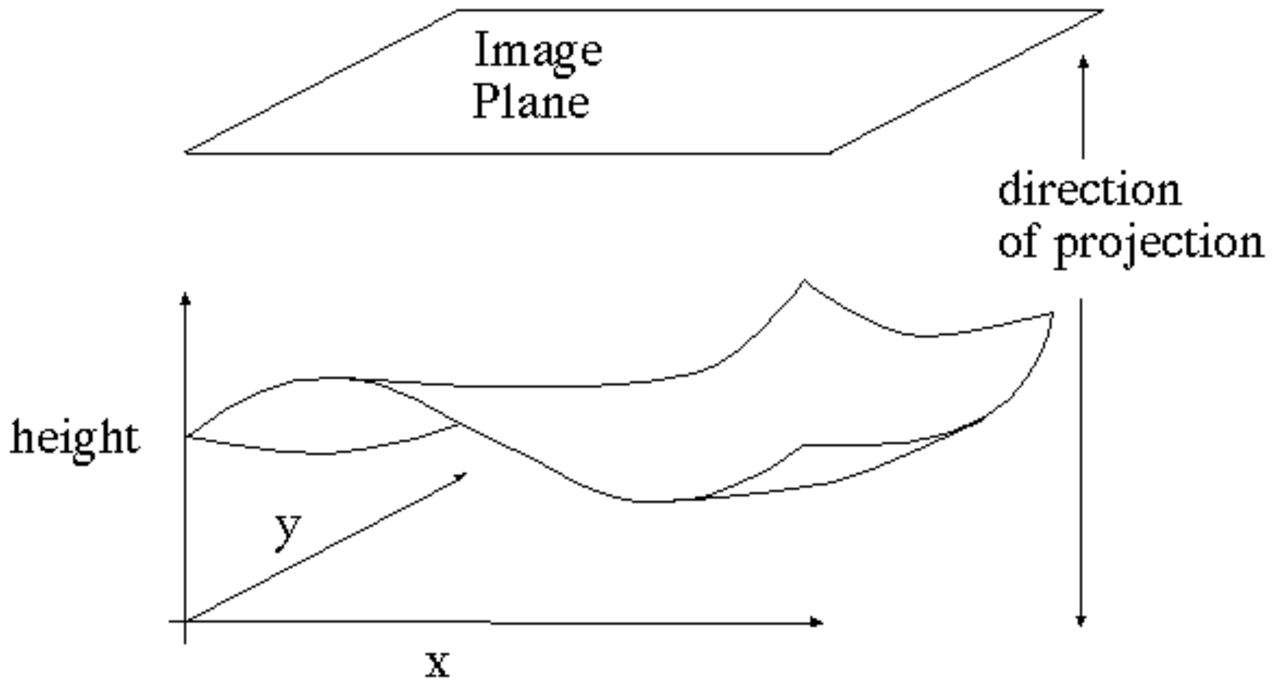
Recovered surface model



F&P 2nd ed., sec. 2.2.4

Image model

- **Known:** source vectors S_j and pixel values $I_j(x, y)$
- **Unknown:** surface normal $N(x, y)$ and albedo $\rho(x, y)$



F&P 2nd ed., sec. 2.2.4

Image model

- **Known:** source vectors S_j and pixel values $I_j(x,y)$
- **Unknown:** surface normal $N(x,y)$ and albedo $\rho(x,y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$\begin{aligned} I_j(x,y) &= k\,\rho(x,y)\big(N(x,y)\cdot S_j\big) \\ &= (\rho(x,y)N(x,y))\cdot\big(k\,S_j\big) \\ &= g(x,y)\cdot V_j \end{aligned}$$

F&P 2nd ed., sec. 2.2.4

Least squares problem

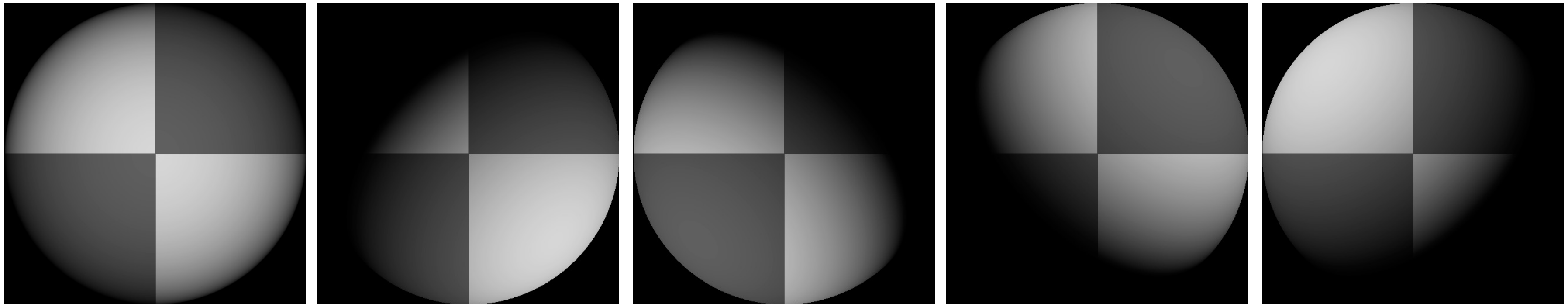
- For each pixel, set up a linear system:

$$\underset{\substack{n\times 3\\ \text{known}}}{\begin{bmatrix} V_1^T \\ V_2^T \\ \vdots \\ V_n^T \end{bmatrix}} \underset{\substack{3\times 1\\ \text{unknown}}}{g(x,y)} = \underset{\substack{n\times 1\\ \text{known}}}{\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{bmatrix}}$$

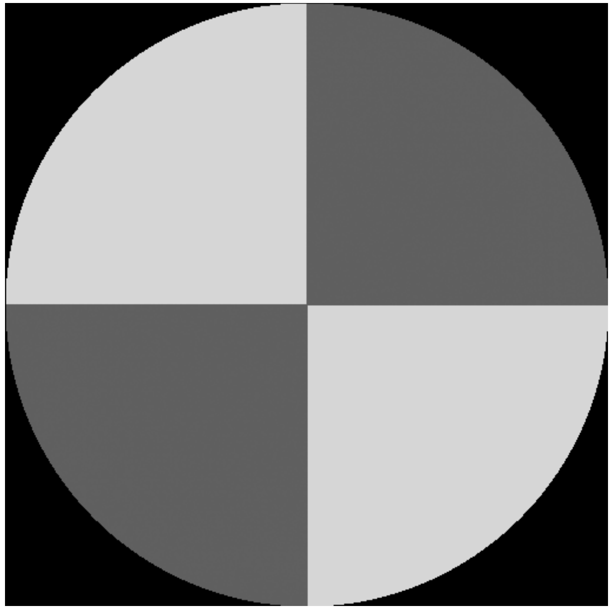
- Obtain least-squares solution for $g(x,y)$, which we defined as $\rho(x,y)N(x,y)$
- Since $N(x,y)$ is the *unit* normal, $\rho(x,y)$ is given by the magnitude of $g(x,y)$
- Finally, $N(x,y) = \frac{1}{\rho(x,y)} g(x,y)$

F&P 2nd ed., sec. 2.2.4

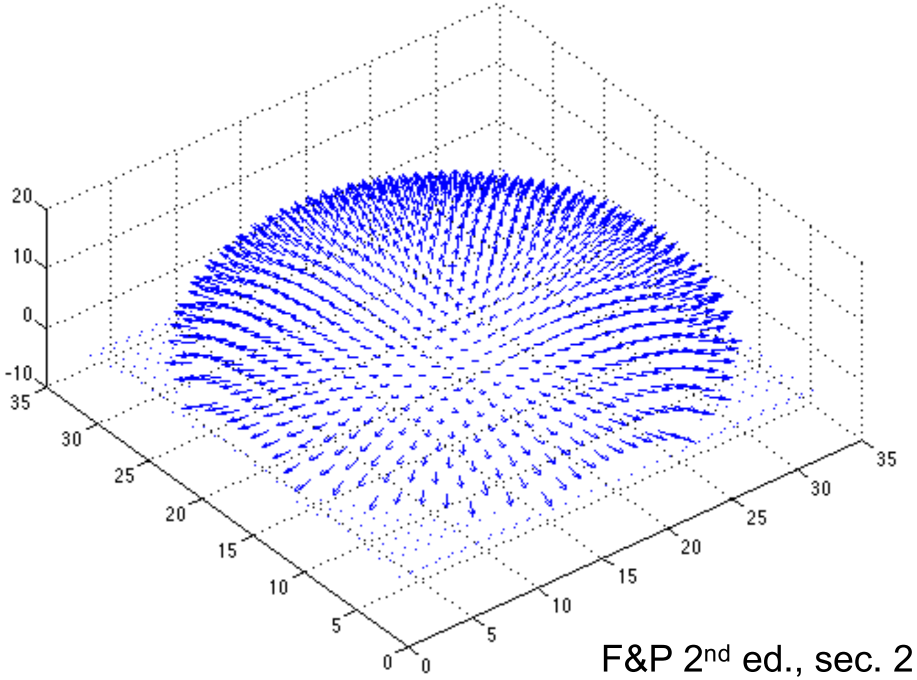
Synthetic example



Recovered albedo



Recovered normal field



F&P 2nd ed., sec. 2.2.4

Recovering a surface from normals

- Recall: the surface is written as

$$(x, y, f(x, y))$$

- This means the unit normal has the following form:

$$N(x, y) = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} \begin{bmatrix} f_x \\ f_y \\ 1 \end{bmatrix}$$

- Write the estimated vector g as

$$g(x, y) = \begin{bmatrix} g_1(x, y) \\ g_2(x, y) \\ g_3(x, y) \end{bmatrix}$$

- Then we obtain values for the partial derivatives of the surface:

$$f_x(x, y) = \frac{g_1(x, y)}{g_3(x, y)}$$

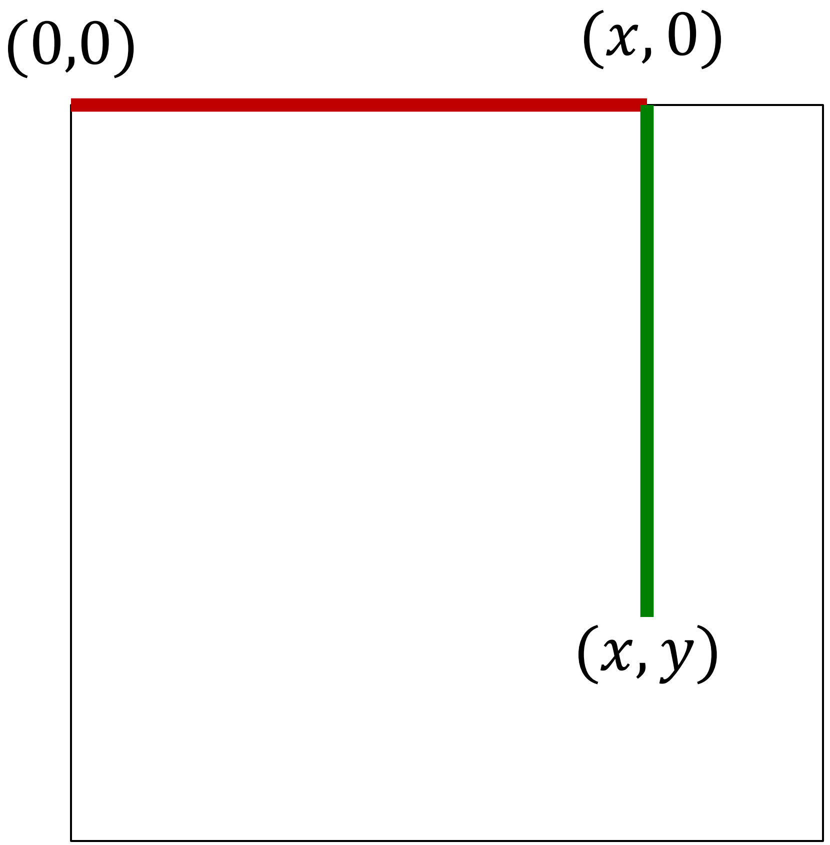
$$f_y(x, y) = \frac{g_2(x, y)}{g_3(x, y)}$$

F&P 2nd ed., sec. 2.2.4

Recovering a surface from normals

- We can now recover the surface height at any point by integration along some path, e.g.

$$f(x,y) =$$
$$\int_0^x f_x(s,0)ds + \int_0^y f_y(x,t)dt + C$$



- For robustness, it is better to take integrals over many different paths and average the results

F&P 2nd ed., sec. 2.2.4

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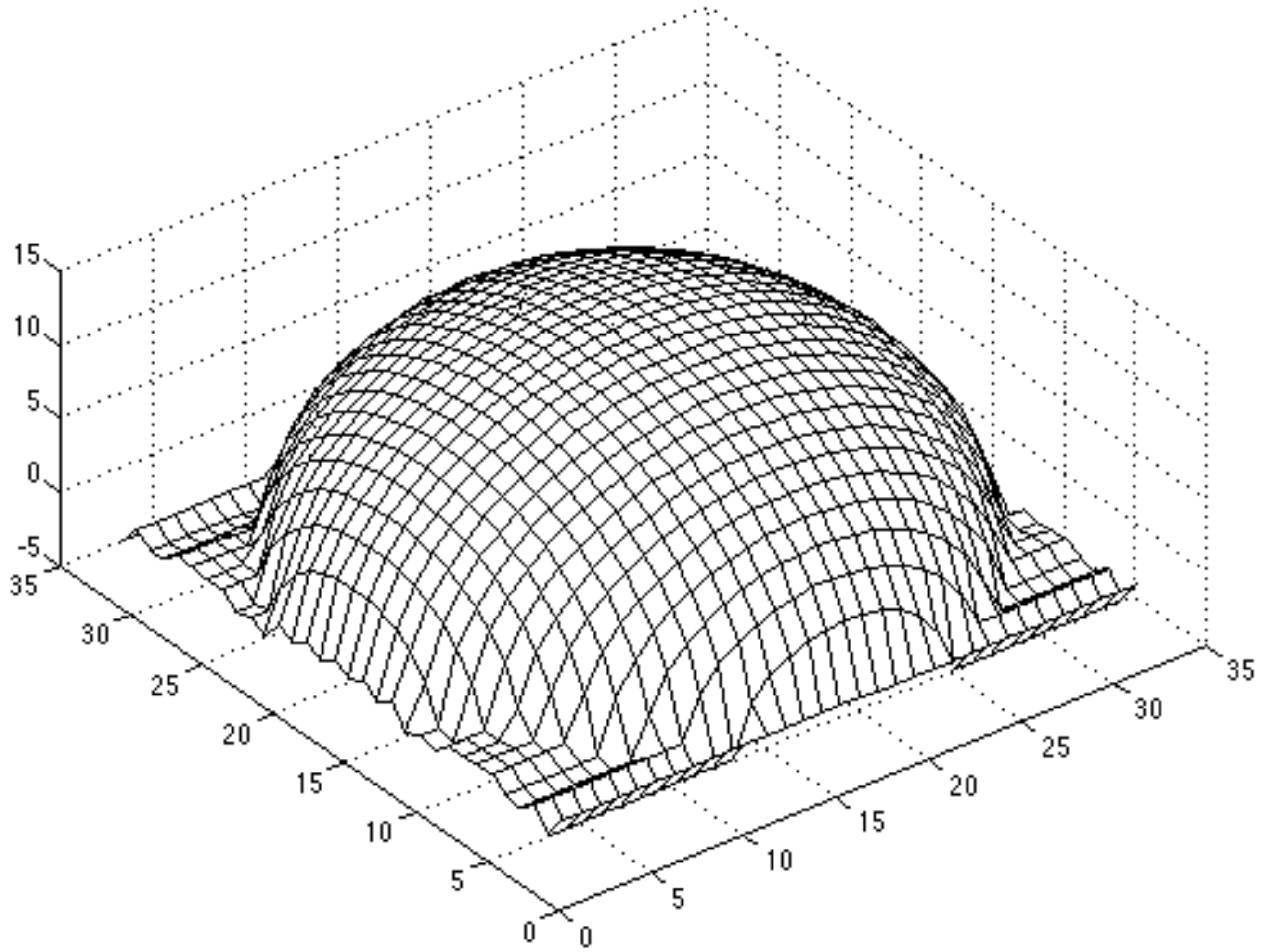
- For robustness, it is better to take integrals over many different paths and average the results

- Note: *integrability* must be satisfied: for the surface f to exist, the mixed second partial derivatives must be equal (or at least similar in practice):

$$\frac{\partial}{\partial y}\left(\frac{g_1(x,y)}{g_3(x,y)}\right) = \frac{\partial}{\partial x}\left(\frac{g_2(x,y)}{g_3(x,y)}\right)$$

F&P 2nd ed., sec. 2.2.4

Surface recovered by integration



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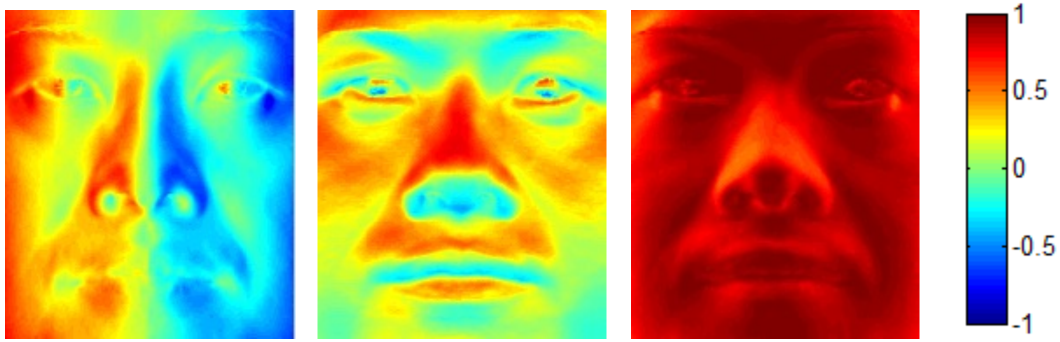
Example

Input

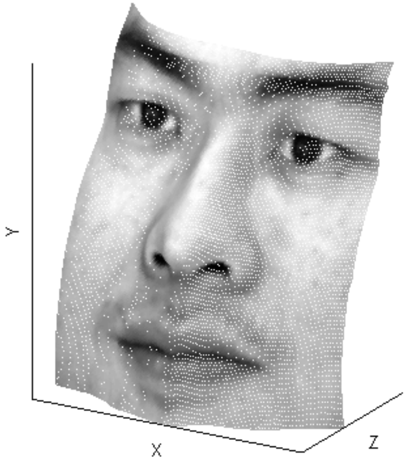
Recovered albedo

Recovered normal field



x y z

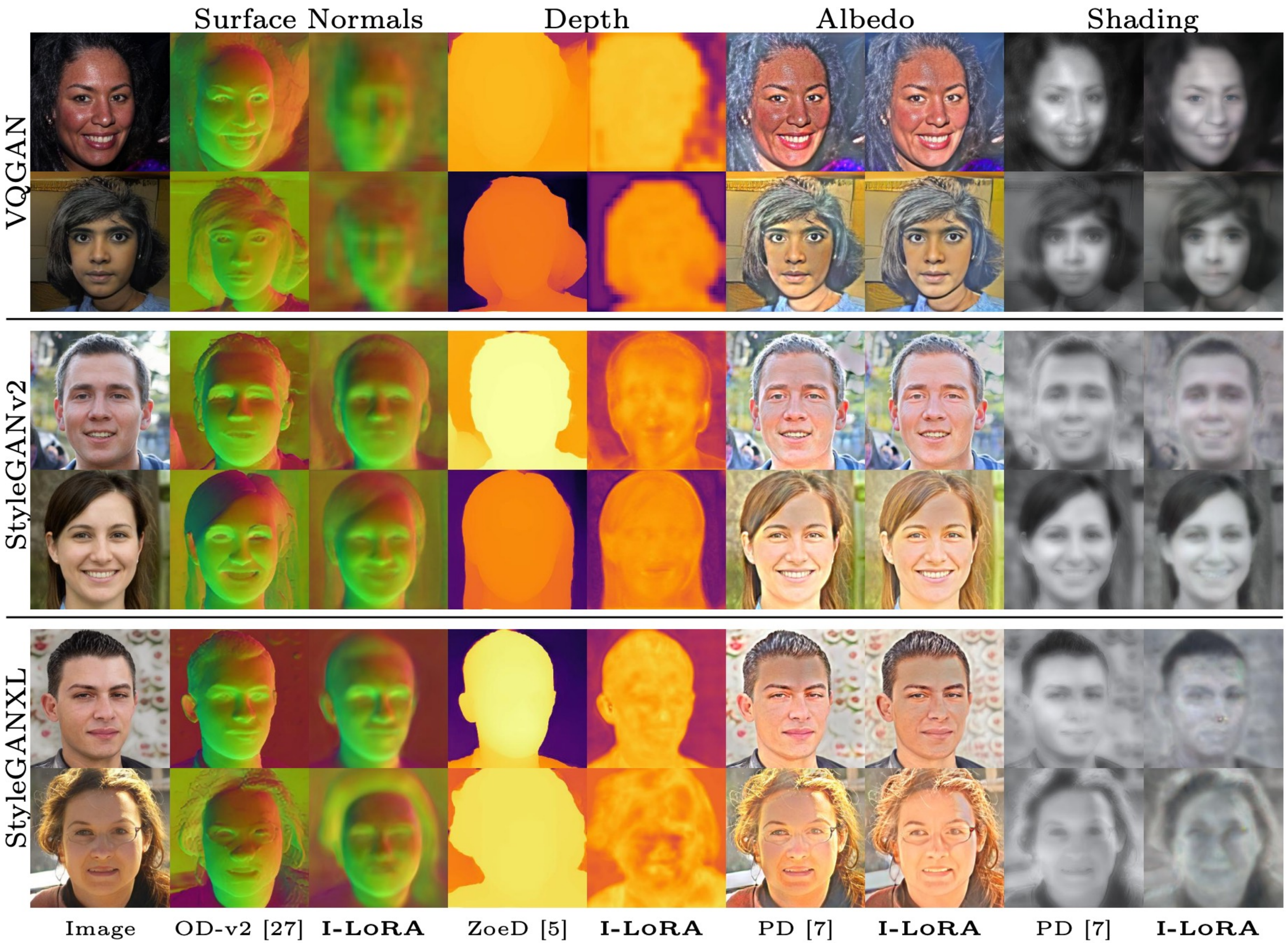
Recovered surface model



Limitations of basic shape from shading

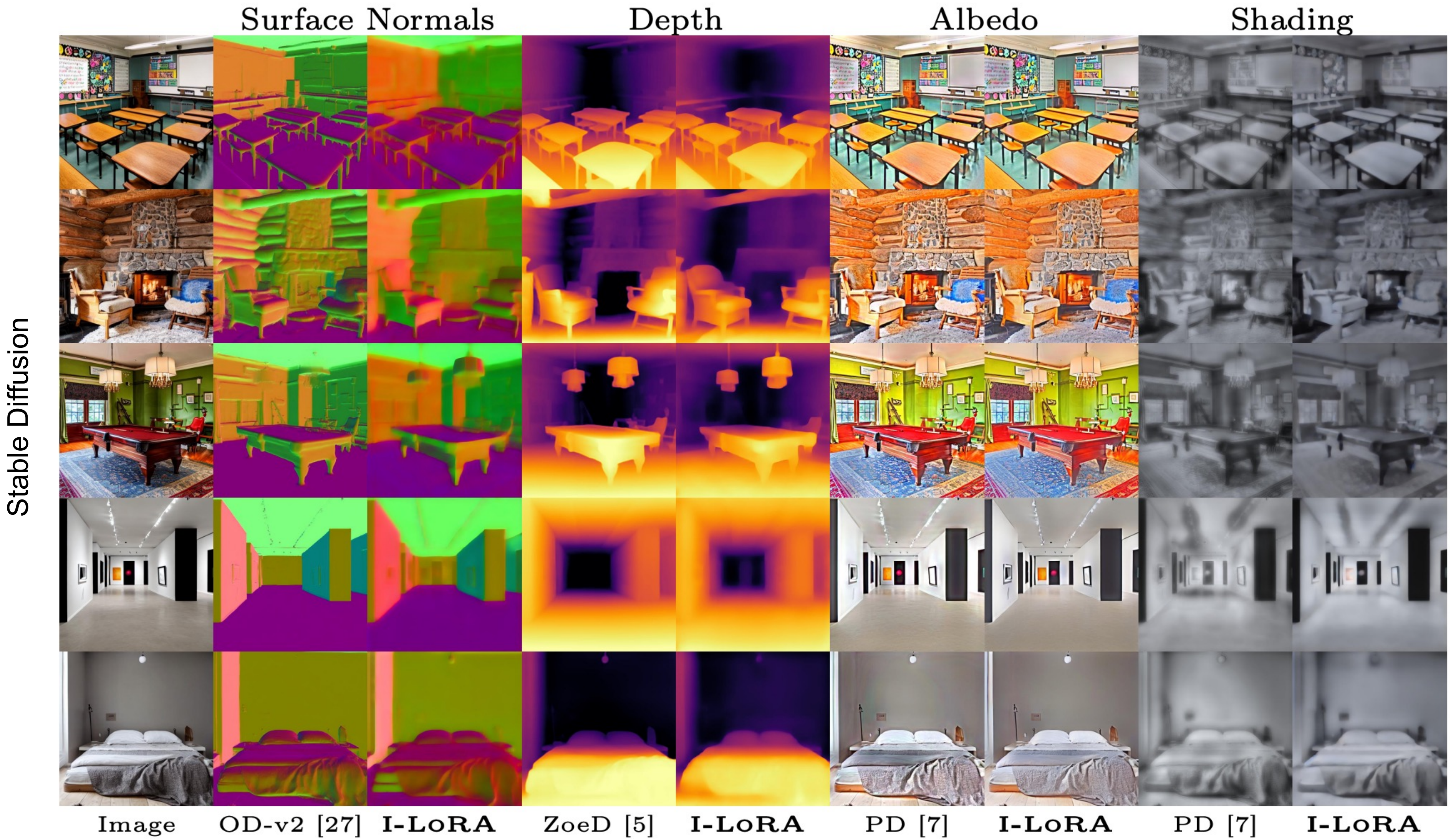
- Orthographic camera model
- Simplistic reflectance and lighting model
- No shadows
- No interreflections
- No missing data
- Integration is tricky

Shape from shading today



X. Du et al. [Generative Models: What do they know? Do they know things? Let's find out!]() CVPR 2024 workshops

Shape from shading today

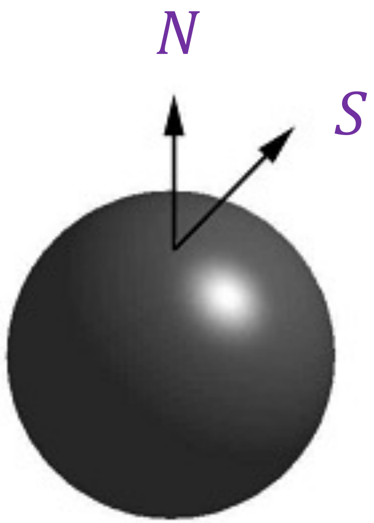


X. Du et al. [Generative Models: What do they know? Do they know things? Let's find out!](#) CVPR 2024 workshops

estimating direction of light sources

Finding the direction of the light source

$$\boxed{I(x,y) = N(x,y) \cdot S(x,y)}$$



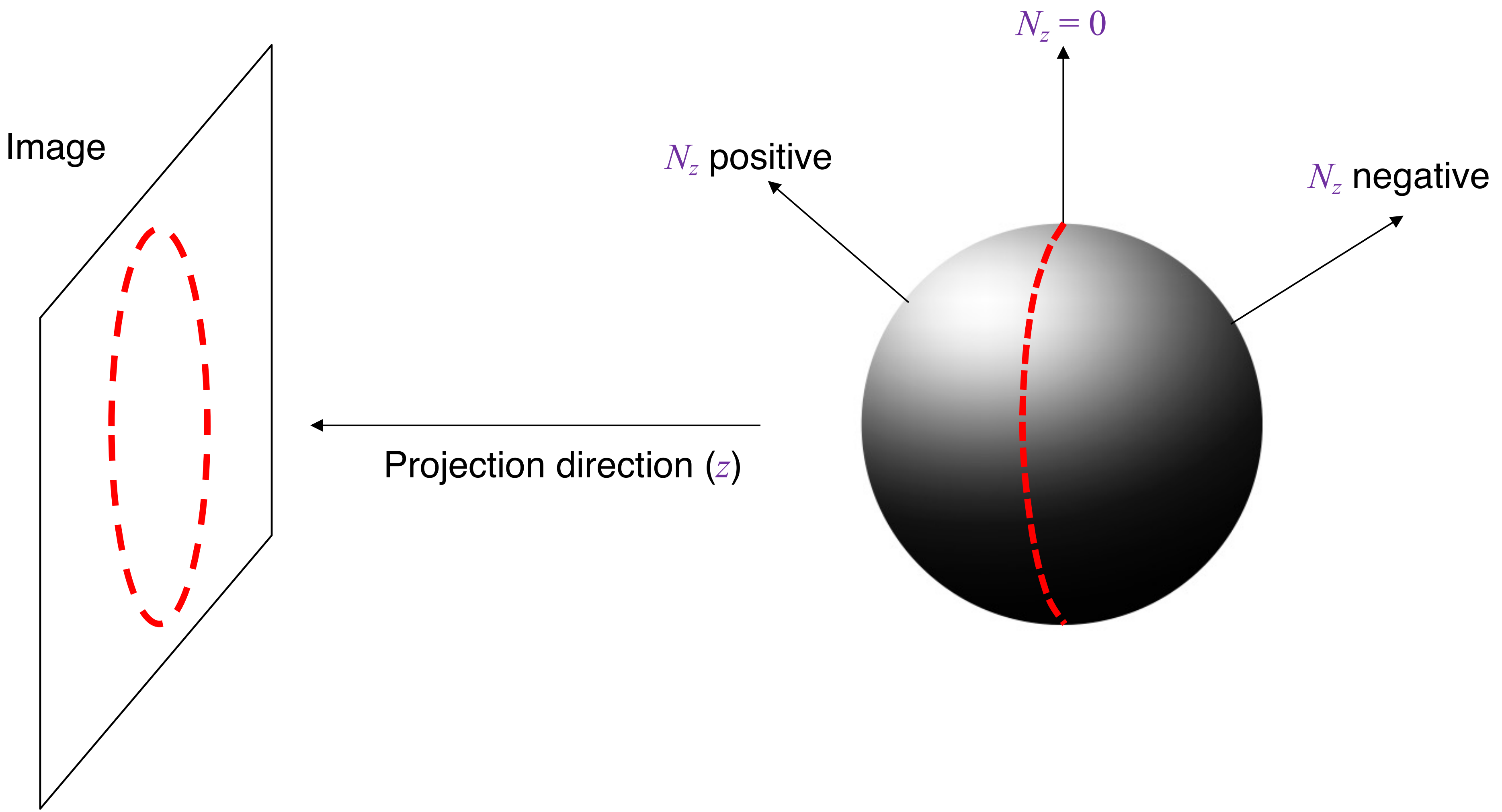
- Full 3D case:

$$\begin{bmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{bmatrix} \begin{bmatrix} S_x \\ S_y \\ S_z \end{bmatrix} = \begin{bmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{bmatrix}$$

P. Nillius and J.-O. Eklundh. Automatic estimation of the projected light source direction. CVPR 2001

Finding the direction of the light source

Consider points on the *occluding contour*:



P. Nillius and J.-O. Eklundh. Automatic estimation of the projected light source direction. CVPR 2001

Finding the direction of the light source

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- For points on the occluding contour ($N_z = 0$):

$$\begin{bmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) \\ \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) \end{bmatrix} \begin{bmatrix} S_x \\ S_y \end{bmatrix} = \begin{bmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{bmatrix}$$

N S

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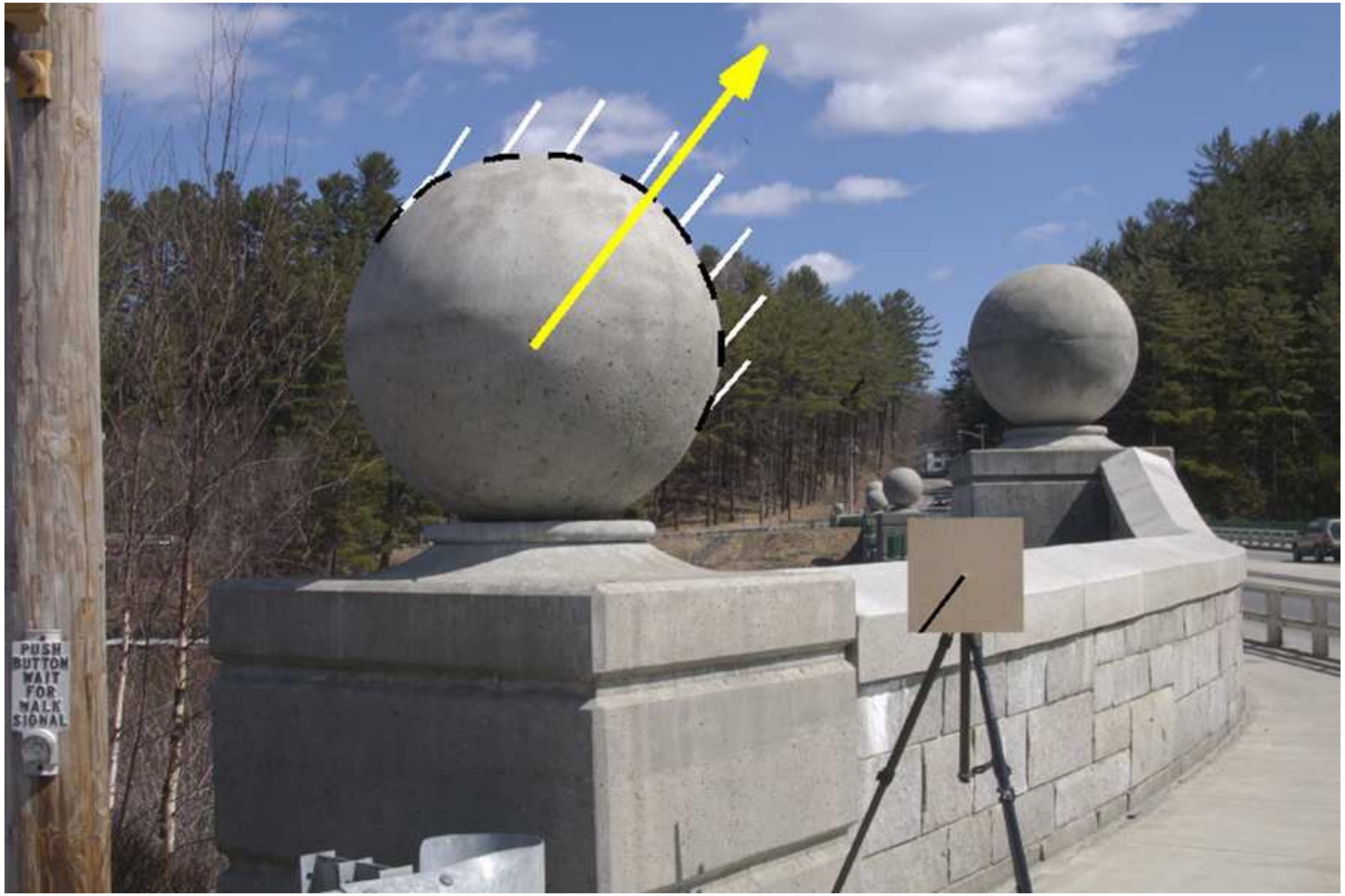
P. Nillius and J.-O. Eklundh. Automatic estimation of the projected light source direction. CVPR 2001

Finding the direction of the light source



P. Nillius and J.-O. Eklundh. [Automatic estimation of the projected light source direction](). CVPR 2001

Application: Detecting composite photos

Real photo

Fake photo

M. K. Johnson and H. Farid. Exposing Digital Forgeries by Detecting Inconsistencies in Lighting.
ACM Multimedia and Security Workshop, 2005

DeepFake detection today

Generated Image | Shadow Errors | Detected Shadow Errors | Vanishing Point Errors | Detected Perspective Errors

A. Sarkar et al. [Shadows Don't Lie and Lines Can't Bend! Generative Models don't know Projective Geometry ... for now](). CVPR 2024

Summary

- Small taste of radiometry
- Reflectance properties of surfaces
- Diffuse and specular reflection
- Shape from shading
- Estimating direction of light sources